

Grade 12
Pre-Calculus Mathematics
Achievement Test

Marking Guide

June 2024

Grade 12 pre-calculus mathematics achievement test.
Marking guide. June 2024

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While the department is committed to making its publications as accessible as possible, some parts of this document are not fully accessible at this time.

Available in alternate formats upon request.

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General Marking Instructions

Please do not make any marks in the student test booklets. If the booklets have marks in them, the marks will need to be removed by departmental staff prior to sample marking should the booklet be selected.

Please ensure that

- the booklet number and the number on the *Answer/Scoring Sheet* are identical
- **students and markers use only a pencil to complete the *Answer/Scoring Sheets***
- the totals of each of the four parts are written at the bottom
- each student's final result is recorded, by booklet number, on the corresponding *Answer/Scoring Sheet*
- the *Answer/Scoring Sheet* is complete
- a photocopy has been made for school records

Once marking is completed, please forward the *Answer/Scoring Sheets* to Manitoba Education and Early Childhood Learning in the envelope provided (for more information see the administration manual).

Marking the Test Questions

The test is composed of constructed response questions and selected response questions. Constructed response questions are worth 1 to 5 marks each, and selected response questions are worth 1 mark each. An answer key for the selected response questions can be found at the beginning of the section "Booklet 2 Questions."

To receive full marks, a student's response must be complete and correct. Where alternative answering methods are possible, the *Marking Guide* attempts to address the most common solutions. For general guidelines regarding the scoring of students' responses, see Appendix A.

Irregularities in Provincial Tests

During the administration of provincial tests, supervising teachers may encounter irregularities. Markers may also encounter irregularities during local marking sessions. Appendix B provides examples of such irregularities as well as procedures to follow to report irregularities.

If an *Answer/Scoring Sheet* is marked with "0" only (e.g., student was present but did not attempt any questions), please document this on the *Irregular Test Booklet Report*.

Assistance

If, during marking, any marking issue arises that cannot be resolved locally, please call Manitoba Education and Early Childhood Learning at the earliest opportunity to advise us of the situation and seek assistance if necessary.

You must contact the Assessment Consultant responsible for this project before making any modifications to the answer keys or scoring rubrics.

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Communication Errors

The marks allocated to questions are primarily based on the concepts and procedures associated with the learning outcomes in the curriculum. For each question, shade in the circle on the *Answer/Scoring Sheet* that represents the marks given based on the concepts and procedures. A total of these marks will provide the preliminary mark.

Errors that are not related to concepts or procedures are called “Communication Errors” (see Appendix A) and will be tracked on the *Answer/Scoring Sheet* in a separate section. There is a $\frac{1}{2}$ mark deduction for each type of communication error committed, regardless of the number of errors per type (i.e., committing a second error for any type will not further affect a student’s mark), with a maximum deduction of 5 marks from the total test mark.

When a given response includes multiple types of communication errors, deductions are indicated in the order in which the errors occur in the response. No communication errors are recorded for work that has not been awarded marks. The total deduction may not exceed the marks awarded.

The student’s final mark is determined by subtracting the communication errors from the preliminary mark.

Example: A student has a preliminary mark of 72. The student committed two E1 errors ($\frac{1}{2}$ mark deduction), four E7 errors ($\frac{1}{2}$ mark deduction), and one E8 error ($\frac{1}{2}$ mark deduction). Although seven communication errors were committed in total, there is a deduction of only $1\frac{1}{2}$ marks.

COMMUNICATION ERRORS / ERREURS DE COMMUNICATION				
Shade in the circles below for a maximum total deduction of 5 marks ($\frac{1}{2}$ mark deduction per error).				
Noircir les cercles ci-dessous pour une déduction maximale totale de 5 points (déduction de 0,5 point par erreur).				
E1 ●	E2 ○	E3 ○	E4 ○	E5 ○
E6 ○	E7 ●	E8 ●	E9 ○	E10 ○

Example: Marks assigned to the student

Marks Awarded	Booklet 1	Selected Response	Booklet 2	Communication Erros (Deduct)	Total
	25	7	40	$1\frac{1}{2}$	$70\frac{1}{2}$
Total Marks	36	9	45	Maximum deduction of 5 marks	90

Scoring Guidelines for Booklet 1 Questions



Question 1

P1

A soccer team consists of 14 players. Determine the number of ways a captain and an assistant captain can be selected.

Solution

$${}_{14}P_2 = 182$$

or

$$\underline{14} \cdot \underline{13} = 182$$

1 mark

Exemplar 1

$${}_{14}C_1 \cdot {}_{13}C_1$$

$$= 182 \text{ ways}$$

1 out of 1

Exemplar 2

$${}_{14}P_1 = 14$$

$${}_{13}P_1 = 13$$

$$\underline{\hspace{1cm}}$$
$$27 \text{ ways}$$

0 out of 1

Question 2

R10

The pH of a person's blood can be found using the formula

$$\text{pH} = 6.1 + \log\left(\frac{B}{C}\right)$$

where B represents the concentration of bicarbonate in the blood, in mEq/L

C represents the concentration of carbonic acid in the blood, in mEq/L

Kansas has a blood pH of 7.41. In her blood sample, the concentration of carbonic acid is 1.41 mEq/L. Determine the concentration of bicarbonate in her blood.

Express the answer correct to one decimal place.

Solution

Method 1

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$10^{1.31} = \frac{B}{1.41} \quad 1 \text{ mark for exponential form}$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Method 2

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$1.31 = \log B - \log 1.41 \quad 1 \text{ mark for quotient law}$$

$$1.31 + \log 1.41 = \log B$$

$$1.4592\dots = \log B$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Exemplar 1

$$pH = 6.1 + \log\left(\frac{B}{C}\right)$$

$$-6.1 \quad 7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) - 6.1$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$10^{1.31} = \frac{B}{1.41}$$

$$1.41(10^{1.31}) = B$$

$$B = 28.789$$

2 out of 2

award full marks (method 1)

E5 (units of measure omitted in final answer)

Exemplar 2

$$= pH = 6.1 + \log\left(\frac{B}{C}\right)$$

$$= \frac{7.41}{6.1} = \frac{6.1}{6.1} + \log\left(\frac{B}{1.41}\right)$$

$$1.2148 = \log\left(\frac{B}{1.41}\right)$$

$$10^{1.2148} = \left(\frac{B}{1.41}\right)$$

1 out of 2

+ ½ mark for substitution

+ 1 mark for exponential form

– ½ mark for procedural error in line 2

E7 (notation error, in line 1 and 2)

Exemplar 3

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right)$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$1.31 = \log B - \log 1.41$$

$$1.31 + \log 1.41 = \log B$$

$$\log 1.31 + \log 1.41 = \log B$$

$$\log(1.31 \cdot 1.41) = \log B$$

$$\log(1.8471) = \log B$$

$$0.266490408 = \log B$$

$$10^{0.266490408} = B$$

$$B = 1.847099999$$

$$B = 1.8 \text{ mEq/L}$$

1 out of 2

award full marks (method 2)

– 1 mark for concept error in line 5 ($\log 1.31$)

E7 (notation error, in lines 1 and 3)

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Solve $(2 \tan x - 1)(\tan x + 1) = 0$ where $x \in \mathbb{R}$.

Solution

Method 1

$$2 \tan x - 1 = 0$$

$$\tan x + 1 = 0$$

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\,647\dots$$

$$x = 0.464$$

$$x = \frac{3\pi}{4}$$

$$x = 3.605$$

$$x = \frac{7\pi}{4}$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

$$\left. \begin{array}{l} x = 0.464 + 2k\pi \\ x = 3.605 + 2k\pi \\ x = \frac{3\pi}{4} + 2k\pi \\ x = \frac{7\pi}{4} + 2k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Method 2

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\,647\dots$$

$$x = \frac{3\pi}{4}$$

2 marks for solving for x (1 mark for each branch)

$$x = 0.464$$

$$\left. \begin{array}{l} x = 0.464 + k\pi \\ x = \frac{3\pi}{4} + k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Exemplar 1

$$\frac{s}{t} \mid \frac{a}{c}$$

$$2 \tan x - 1 = 0$$

$$\tan x = \frac{1}{2}$$

$$x = \tan^{-1}\left(\frac{1}{2}\right)$$

$$x = 26.6^\circ$$

$$Q1, 3$$

$$26.6^\circ, 333.4^\circ$$

$$\tan x + 1 = 0$$

$$\tan x = -1$$

$$x = \tan^{-1}(-1)$$

$$x = -45^\circ$$

$\tan$ is negative

$$Q2, 4$$

$$135^\circ, 315^\circ$$

$$\frac{s}{x} \mid \frac{a}{c}$$

1½ out of 3

+ 1½ marks for solving for x . (three correct values of x)

E6 (rounding error)

E7 (notation error in line 6)

Exemplar 2

$$2 \tan x - 1$$

$$\tan x + 1$$

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$\theta_R = 26.6^\circ$$

$$\theta_R = 45^\circ$$

$$\theta_1 = 0.46$$

$$\theta_2 = 0.46 + 2\pi k, k \in \mathbb{Z}$$

$$\theta_3 = 0.79 + 2\pi k, k \in \mathbb{Z}$$

1½ out of 3

+ ½ mark for solving for x . (one correct value of x)

+ 1 mark for general solution

E2 (changing an equation to an expression in line 1)

E6 (rounding error)

E3 (variable introduced without being defined)

Exemplar 3

$$2\tan x - 1 = 0$$

$$2\tan x = 1$$

$$\tan x = \frac{1}{2}$$

$$x = 0.463647609$$

$$\tan x + 1 = 0$$

$$\tan x = -1$$

$$x = -0.785398163$$

1 out of 3

+ 1 mark for solving for x (two correct values of x)

E7 (notation error in line 3)

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Determine and simplify the 7th term in the binomial expansion of $\left(\frac{2}{x^2} + 3x\right)^9$.

Solution

$$t_7 = {}_9C_6 \left(\frac{2}{x^2}\right)^3 (3x)^6 \quad 2 \text{ marks (1 mark for } {}_9C_6; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 84 \left(\frac{8}{x^6}\right) (729x^6)$$

$$= 489\,888$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Exemplar 1

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$t_{6+1} = {}_9C_6 a^3 b^6$$

$$t_7 = 84 \left(\frac{2}{x^2} \right)^3 (3x)^6$$

$$t_7 = 84 \left(\frac{8}{x^5} \right) (729x^6)$$

$$t_7 = \frac{489888x^6}{x^5}$$

$$t_7 = 489888x$$

2½ out of 3

+ 2 marks (1 mark for ${}_9C_6$; ½ mark for each consistent factor)

+ ½ mark for simplification of coefficient

Exemplar 2

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$t_7 = {}_9C_6 \frac{2}{x^2}^{9-6} \cdot 3x^6$$

$$t_7 = \left(\frac{9!}{6!(9-6)!} \right) \cdot \frac{8}{x^6} \cdot 3x^6$$

$$t_7 = 84 \cdot \frac{8}{x^6} \cdot 3x^6$$

$$= \frac{672}{x^6} \cdot 3x^6$$

$$= \frac{2016x^6}{x^6} = 2016$$

$$t_7 = 2016$$

2½ out of 3

+ 2 marks (1 mark for ${}_9C_6$; ½ mark for each consistent factor)

+ ½ mark for simplification of exponent

E4 (missing brackets but still implied in line 2)

Exemplar 3

$$t_{6+1} = {}_9C_6 \left(\left(\frac{2}{x^2} \right)^{9-6} \right) (3x)^6$$

$$t_7 = 84 \left(\left(\frac{2}{x^2} \right)^3 \right) (3x)^6$$

$$t_7 = 84 (x^2)^3 (x^6)$$

$$t_7 = 84 (x^6) (x^6)$$

$$t_7 = 84 (x^6)^2$$

$$t_7 = 84x^{12}$$

2 out of 3

+ 2 marks (1 mark for ${}_9C_6$; $\frac{1}{2}$ mark for each consistent factor)

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Question 5

T1

A winch, with a diameter of 14 cm, is used to pull a boat out of the water. As the winch rotates, it pulls in a cable. Determine the length of cable that is pulled in if the winch rotates 526° .



Solution

$$\theta = (526) \left(\frac{\pi}{180} \right)$$

1 mark for conversion

$$\theta = \frac{263\pi}{90}$$

$$\theta = 9.180\,431\dots$$

$$s = \theta r$$

$$s = \left(\frac{263\pi}{90} \right) (7)$$

1 mark for substitution

$$s = \frac{1841\pi}{90} \text{ cm}$$

2 marks

or

$$s = 64.263 \text{ cm}$$

Exemplar 1

$$S = \theta r$$

$$S = ?$$

$$\theta = 526^\circ = 526 \times \frac{\pi}{180^\circ} = 2.922\pi$$

$$r = 7\text{cm}$$

$$S = 2.922(7\text{cm})$$

$$S = 20.456\text{cm}$$

2 out of 2

award full marks

E7 (transcription error in line 5)

Exemplar 2

$$S = \theta r$$

$$S = \left(\frac{526\pi}{180}\right)(14\text{cm})$$

$$S = 128.526$$

$$\frac{526}{1} \times \frac{\pi}{180} = \frac{526\pi}{180}$$

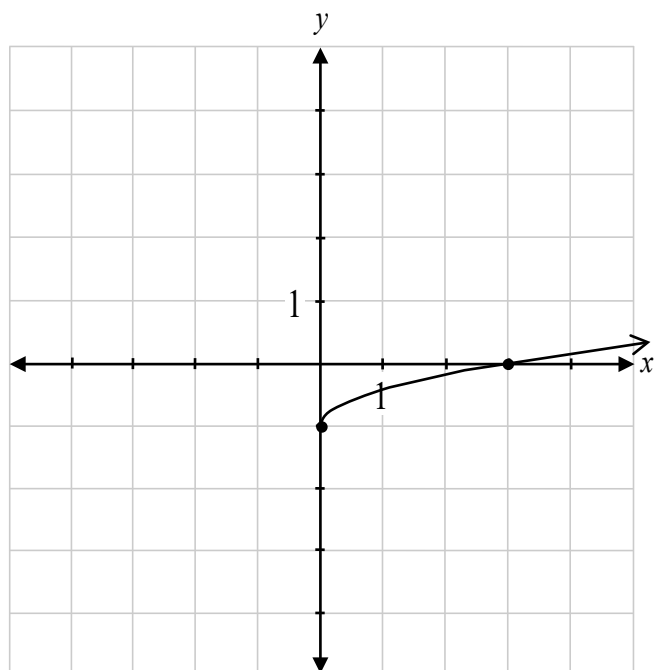
1½ out of 2

award full marks

– ½ mark for procedural error in line 2

E5 (units of measure omitted in final answer)

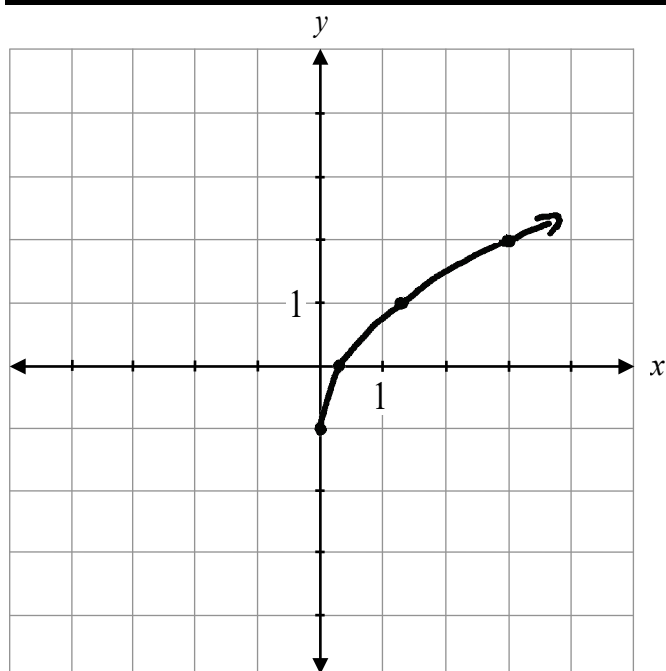
Sketch the graph of $f(x) = \sqrt{\frac{1}{3}x} - 1$.

Solution

1 mark for shape of a radical function
1 mark for horizontal stretch
1 mark for vertical translation

3 marks

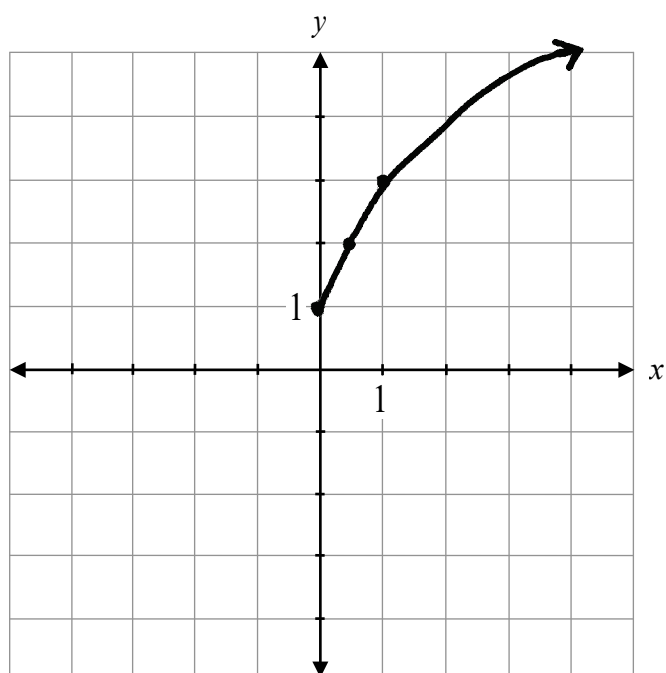
Exemplar 1



2 out of 3

- + 1 mark for shape of a radical function
- + 1 mark for vertical translation

Exemplar 2



1 out of 3

- + 1 mark for shape of a radical function

Question 7

R2, R3, R5

Given $y = f(x)$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection over the y -axis
- vertical stretch by a factor of 3
- horizontal translation 2 units left

Solution

$$g(x) = \underline{3f(-(x+2))}$$

1 mark for horizontal reflection

1 mark for vertical stretch

1 mark for horizontal translation

3 marks

Exemplar 1

$$g(x) = \underline{3g(-x+2)}$$

1½ out of 3

+ 1 mark for horizontal reflection

+ 1 mark for vertical stretch

– ½ mark for procedural error (using “g” instead of “f”)

Exemplar 2

$$g(x) = \underline{3f(-x+2)}$$

2 out of 3

+ 1 mark for horizontal reflection

+ 1 mark for vertical stretch

Exemplar 3

$$g(x) = \underline{3f-(x+2)}$$

2½ out of 3

award full marks

– ½ mark for procedural error (omitting brackets)

Exemplar 4

$$g(x) = \underline{-3(x-2)}$$

0 out of 3

+ 1 mark for vertical stretch

– 1 mark for concept error (missing “f”)

Explain why the graph of $y = \frac{6x+7}{3x+5}$ has a horizontal asymptote at $y = 2$.

Solution

When the degree of the numerator and the denominator are the same, the horizontal asymptote is $y = \frac{a}{b}$ where a is the leading coefficient in the numerator and b is the leading coefficient in the denominator.

1 mark

Exemplar 1

You divide the coefficient of the numerator by the coefficient of the denominator.

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

Exemplar 2

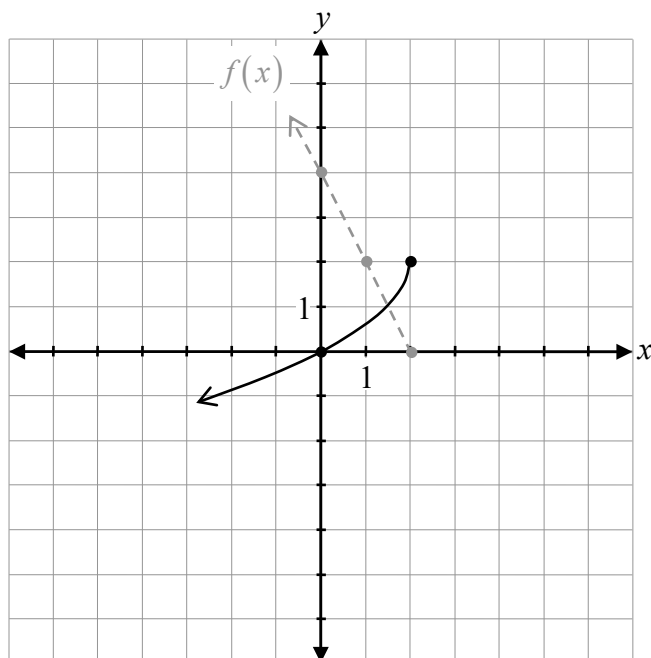
The degrees on the top and bottom are equal
∴ the leading coefficient tells you where the H.A is.

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

Given the graph of $y = f(x)$, sketch the graph of $y = -\sqrt{f(x)} + 2$.

Solution

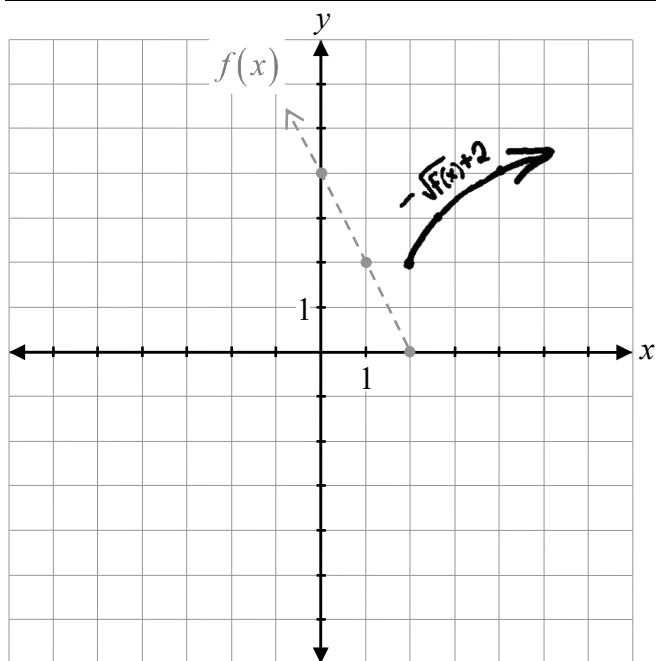
1 mark for shape of $\sqrt{f(x)}$

1 mark for vertical reflection

1 mark for vertical translation

3 marks

Exemplar 1

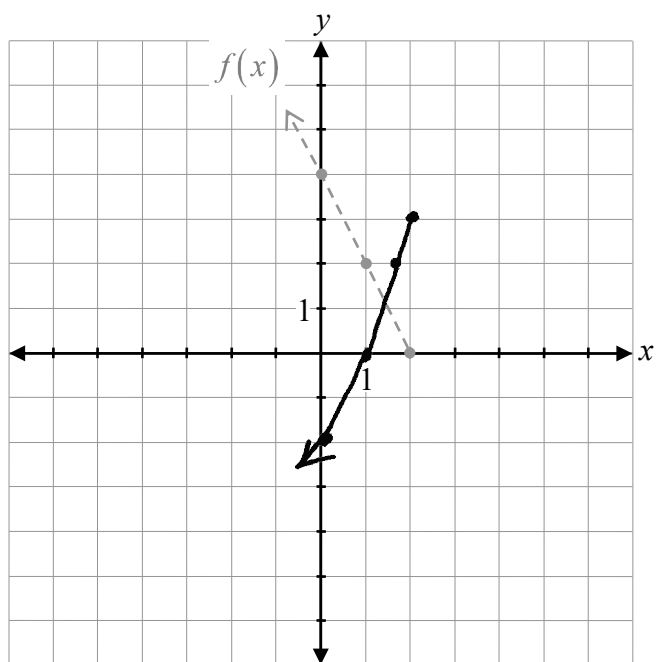


2 out of 3

+ 1 mark for shape of $\sqrt{f(x)}$

+ 1 mark for vertical translation

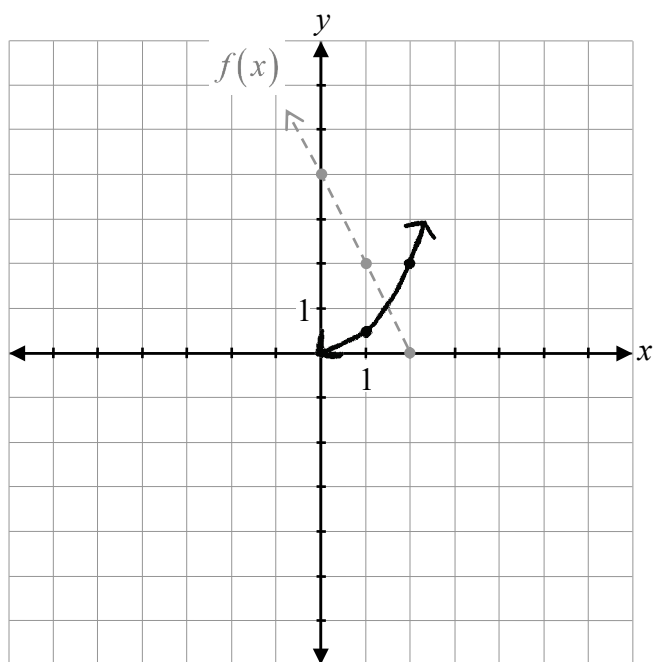
Exemplar 2



1 out of 3

+ 1 mark for vertical reflection

Exemplar 3



2 out of 3

- + 1 mark for vertical reflection
- + 1 mark for vertical translation

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Using the laws of logarithms, completely expand the given expression.

$$\log\left(\frac{A}{\sqrt[3]{B} \cdot C^4}\right)$$

Solution

$$\log A - \frac{1}{3}\log B - 4\log C$$

1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

3 marks

Exemplar 1

$$\log A - \frac{1}{3} \log B + 4 \log C$$

2½ out of 3

award full marks

– ½ mark for procedural error (incorrect sign of third term)

Exemplar 2

$$\log A - \sqrt[3]{\log_{10} B} + \log_{10} C^4$$

$$\log_{10} A - \sqrt[3]{\log_{10} B} + 4 \log_{10} C$$

2 out of 3

+ 1 mark for product law

+ ½ mark for power law

+ 1 mark for quotient law

– ½ mark for procedural error (incorrect sign of third term)

Prove the following identity for all permissible values of θ .

$$\cot \theta - \tan \theta = \frac{2 \cos 2\theta}{\sin 2\theta}$$

Solution

Left-Hand Side	Right-Hand Side
$\cot \theta - \tan \theta$ $\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$ $\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$	$\frac{2 \cos 2\theta}{\sin 2\theta}$ $\frac{2 \cos 2\theta}{2 \sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$

1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Exemplar 1

Left-Hand Side	Right-Hand Side
$\cot \theta - \tan \theta$ $\overset{\times \cos \theta}{\frac{\cos \theta}{\sin \theta}} - \frac{\sin \theta}{\overset{\times \sin \theta}{\cos \theta}}$ $\frac{\cos^2 \theta}{\sin \theta \cos \theta} - \frac{\sin^2 \theta}{\sin \theta \cos \theta}$ $\frac{\cancel{\cos^2 \theta} - \cancel{\sin^2 \theta}}{\cancel{\sin \theta} \cos \theta}$ $\cos \theta - \sin \theta$	$\frac{2 \cos 2 \theta}{\sin 2 \theta}$ $\frac{2 \cos 2 \theta}{2 \sin \theta \cos \theta}$ $\frac{2 \cancel{\cos^2 \theta} - \cancel{\sin^2 \theta}}{2 \cancel{\sin \theta} \cos \theta}$ $\cos \theta - \sin \theta$

1 out of 3

+ 1 mark for substitution of appropriate identities

Exemplar 2

Left-Hand Side	Right-Hand Side
$\frac{\cos \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \cdot \frac{\sin \theta}{\sin \theta}$ $= \frac{\cos^2 \theta}{\sin \theta} - \frac{\sin^2 \theta}{\cos \theta}$	$\frac{2 \cos 2 \theta}{\sin 2 \theta}$ $= 2 \frac{(\cos^2 \theta - \sin^2 \theta)}{\sin \theta \cos \theta}$ $= 2 (\cos \theta - \sin \theta)$

0 out of 3

Exemplar 3

Left-Hand Side	Right-Hand Side
$\frac{\cot \theta - \tan \theta}{\cos \theta \cdot \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \cdot \sin \theta}$ $\frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta \sin \theta}$	$\frac{2 \cos 2 \theta}{\sin 2 \theta}$

1 out of 3

+ 1 mark for algebraic strategies

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Question 12

R11

Express $p(x) = -2x^3 + x^2 + 13x + 6$ in completely factored form.

Solution

$$p(-2) = -2(-2)^3 + (-2)^2 + 13(-2) + 6$$

$$p(-2) = -2(-8) + 4 - 26 + 6$$

$$p(-2) = 0$$

1 mark for identifying one zero of $p(x)$

$\therefore (x + 2)$ is a factor.

$$\begin{array}{r|rrrr} -2 & -2 & 1 & 13 & 6 \\ & \downarrow & 4 & -10 & -6 \\ \hline & -2 & 5 & 3 & 0 \end{array}$$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 2)(-2x^2 + 5x + 3)$$

$$p(x) = -(x + 2)(2x^2 - 5x - 3)$$

$$p(x) = \underline{-(x + 2)(2x + 1)(x - 3)}$$

1 mark for consistent product of factors

3 marks

Exemplar 1

$$\pm 1, \pm 2, \pm 3, \pm$$

$$\begin{aligned} & -2(-2)^3 + (-2)^2 + 13(-2) + 6 \\ & = 16 + 4 - 26 + 6 \\ & = 0 \end{aligned}$$

$$\begin{array}{r|rrrr} 2 & -2 & 1 & 13 & 6 \\ & & -4 & 10 & 6 \\ \hline & -2x^2 & 5x & 3 & 0 \end{array}$$

$$\begin{aligned} & -2x^2 + 5x + 3 \\ & (-2x-1)(x-3) \end{aligned}$$

$$\begin{array}{r|l} -6 & 5 \\ \hline 6x-1 & \checkmark \end{array}$$

$$(-2x-1)(x-3)(x+2) = 0$$

$$p(x) = \underline{-\frac{1}{2}, 3, -2}$$

2½ out of 3

award full marks

– ½ mark for procedural error (determining the zeroes)

Exemplar 2

$$\pm 1 \pm 2 \pm 3 \pm 6$$

$$\begin{array}{r|rrrr} 3 & -2 & 1 & 13 & 6 \\ & & -6 & -15 & -6 \\ \hline & -2 & -5 & -2 & 0 \end{array}$$

$$\begin{aligned} & (-2x^2 - 5x - 2)(x-3) \\ & -2x^2 - 4x + (-x-2) \\ & -2x(x+2) - 1(x+2) \\ & (x+2)(-2x-1)(x-3) \end{aligned}$$

$$p(x) = \underline{(x+2)(-2x-1)(x-3)}$$

3 out of 3

Exemplar 3

$$\begin{array}{r} -2x^3 + x^2 + 13x + 6 \\ -2 \overline{) \begin{array}{r} -2 \quad 1 \quad 13 \quad 6 \\ \downarrow \quad 4 \quad -10 \quad -6 \\ -2 \quad 5 \quad 3 \quad 0 \\ -2x^2 + 5x + 3 \end{array}} \\ 2x^2 - 5x - 3 \\ 2x^2 + x - 6x - 3 \\ x(2x+1) - 3(2x+1) \\ (x-3)(2x+1) \end{array}$$

$$p(x) = \underline{(x+2)(x-3)(2x+1)}$$

2½ out of 3

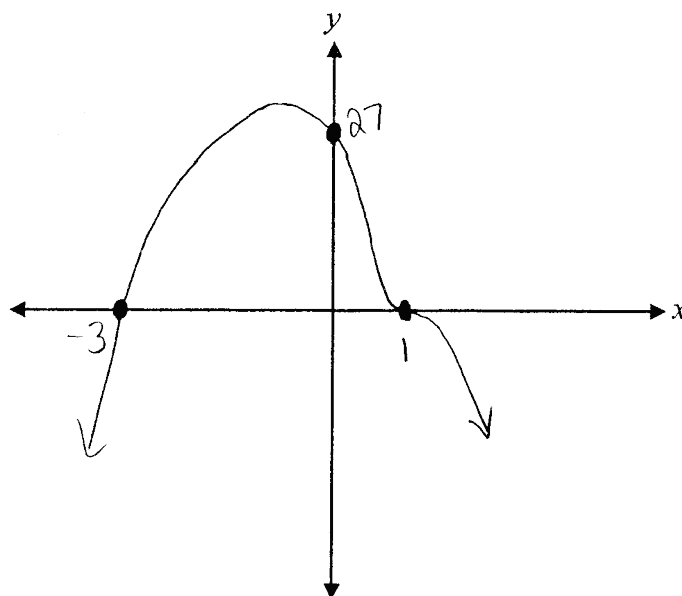
award full marks

– ½ mark for procedural error in line 4

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Kaitlyn was asked to sketch the graph of the polynomial function $p(x) = -(x-1)(x+3)^3$.

Kaitlyn's graph:



Describe an error on her graph.

Solution

Kaitlyn's graph shows a multiplicity of 3 at $x = 1$ instead of at $x = -3$.

1 mark

Exemplar 1

The graph's reaction to the multiplicities is incorrect and she didn't distribute the negative.

0 out of 1

Exemplar 2

The multiplicity of 3
was to be
at $x-3$.

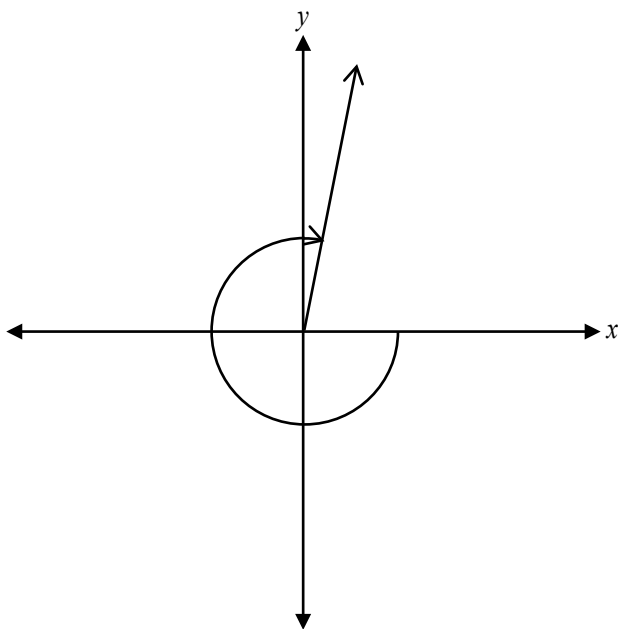
0 out of 1

Exemplar 3

She did not draw the proper behaviour for a multiplicity of 3. At $x=-3$, the graph should "trail" x-axis before crossing. However her graph goes straight through.

1 out of 1

Sketch the angle of -5 radians in standard position.

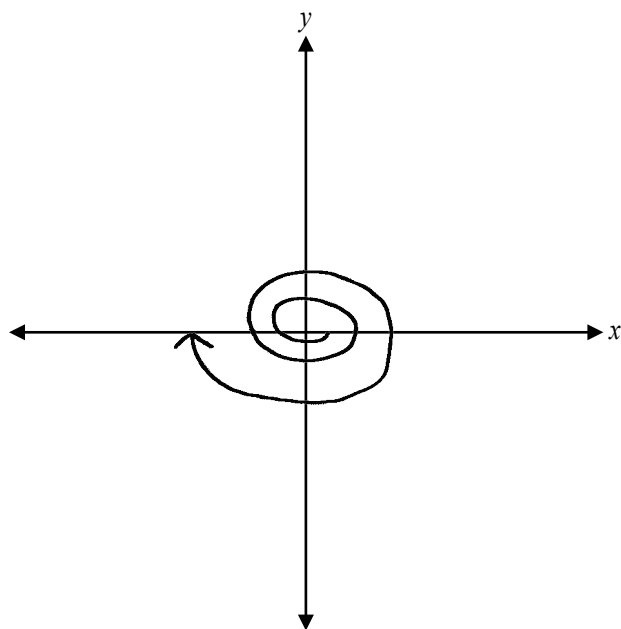
Solution

$\frac{1}{2}$ mark for an appropriate angle in quadrant I

$\frac{1}{2}$ mark for the correct direction of an angle in standard position

1 mark

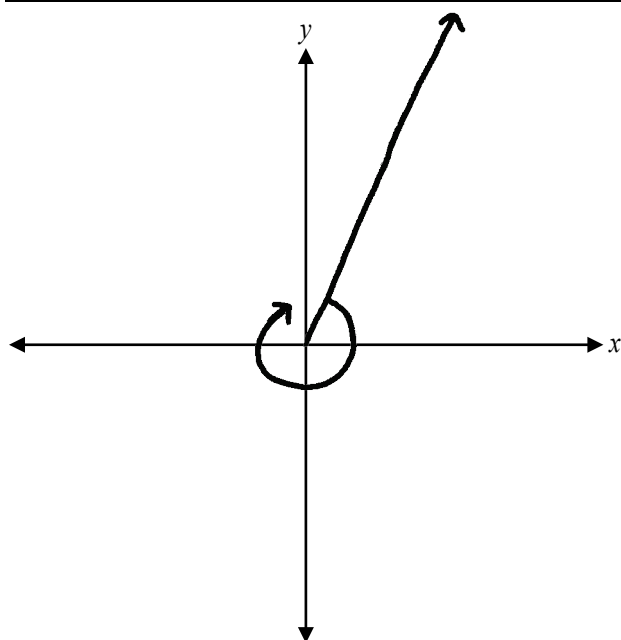
Exemplar 1



½ out of 1

+ ½ mark for the correct direction of an angle in standard position

Exemplar 2



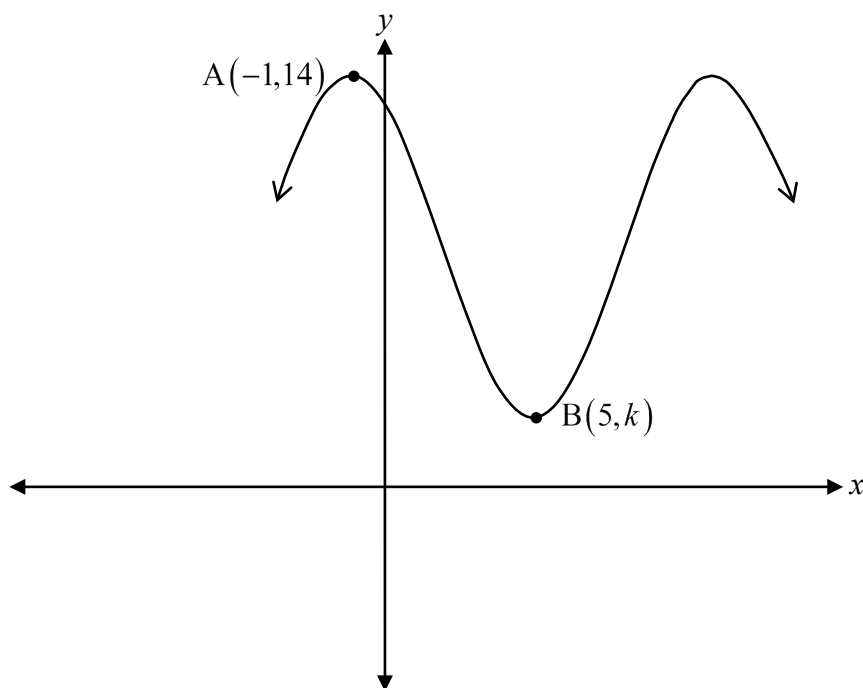
½ out of 1

+ ½ mark for an appropriate angle in quadrant 1

Question 15

T4

The graph of a sinusoidal function is sketched below. The point A is a maximum and the point B is a minimum. State the value of k if the amplitude of the function is 6.



Solution

$$k = 2$$

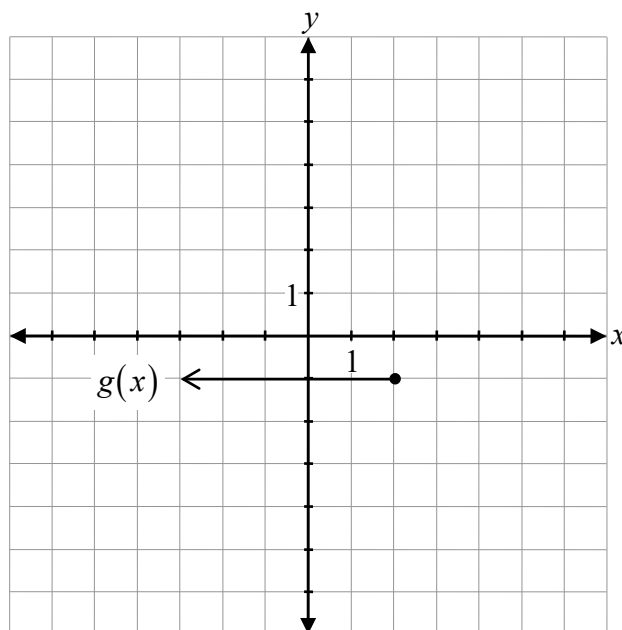
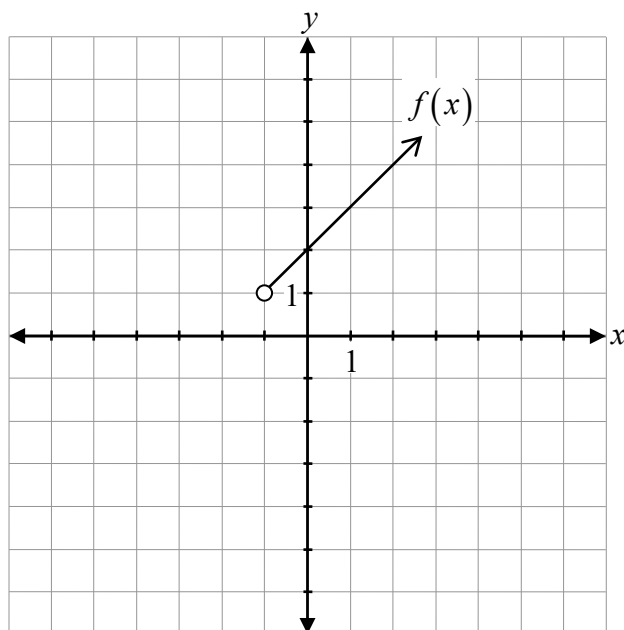
1 mark

Exemplar 1

$$14 - 6 = 8 \quad K = 8$$

0 out of 1

Given the graphs of $f(x)$ and $g(x)$, state the domain of $(f + g)(x)$.

**Solution**

Domain: $\{x \mid -1 < x \leq 2, x \in \mathbb{R}\}$

1 mark

or

Domain: $(-1, 2]$

Exemplar 1

Domain: $-1 < x \leq 2$

0 out of 1

Exemplar 2

Domain: $(-1, 2)$

1 out of 1

award full marks

E8 (bracket error made when stating domain)

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Scoring Guidelines for Booklet 2 Questions

Answer Key for Multiple Choice Questions

Question	Answer	Learning Outcome
17	B	R2, R6
18	C	R12
19	B	T3
20	A	T6
21	C	R10
22	D	P3
23	D	P2
24	B	P4
25	B	R9

Question 17

R2, R6

Given that the point $(-13, 7)$ is on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f^{-1}(x+2)$.

a) $(7, -11)$

b) $(5, -13)$

c) $(7, -15)$

d) $(9, -13)$

Question 18

R12

Given that $x = 0$, $x = 1$, $x = 2$ and $x = -3$ are zeros of the polynomial function $p(x)$, identify a possible equation for $p(x)$.

a) $p(x) = -3x(x-1)(x-2)$

b) $p(x) = (x+1)(x+2)(x-3)^2$

c) $p(x) = x(x+3)^2(x-1)(x-2)$

d) $p(x) = (x-2)(x-1)^3(x+3)$

Question 19

T3

Identify the interval where the angle, θ , could exist if $\cos \theta = -\frac{4}{5}$ and $\csc \theta = \frac{5}{3}$.

a) $\left(0, \frac{\pi}{2}\right)$

b) $\left(\frac{\pi}{2}, \pi\right)$

c) $\left(\pi, \frac{3\pi}{2}\right)$

d) $\left(\frac{3\pi}{2}, 2\pi\right)$

Question 20

T6

Identify a non-permissible value of θ in the identity, $\frac{\cot \theta}{\csc \theta} = \cos \theta$.

a) π

b) $\frac{\pi}{4}$

c) $\frac{3\pi}{2}$

d) $\frac{\pi}{2}$

Question 21

R10

Identify the solution of the equation, $e^{2x} = 16$.

a) $x = \ln 8$

b) $x = 8$

c) $x = \frac{\ln 16}{2}$

d) $x = \frac{\log 16}{2}$

Question 22

P3

A committee of 6 people is to be selected from 90 students. Identify an expression that represents the number of possible committees if two of the students, Evelyn and Wayne, must be on the committee.

a) ${}_{90}C_4$

b) ${}_{90}C_6$

c) ${}_{88}C_6$

d) ${}_{88}C_4$

Question 23

P2

Identify the expression for the number of arrangements of the letters in the word RUSSELL.

a) $\frac{7!}{4!}$

b) $7!$

c) $\frac{7!}{2!}$

d) $\frac{7!}{2!2!}$

Question 24

P4

Given the row of Pascal's triangle below, identify the third value of the next row.

1 7 21 35 35 21 7 1

a) 8

b) 28

c) 56

d) 70

Question 25

R9

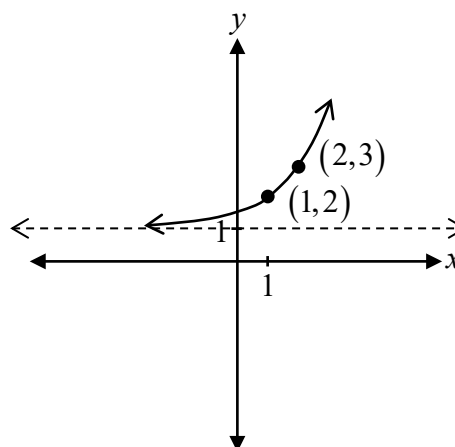
Identify the equation represented by the graph sketched below.

a) $y = \left(\frac{1}{2}\right)^x + 1$

b) $y = 2^{x-1} + 1$

c) $y = 2^{x+1} + 1$

d) $y = \left(\frac{1}{2}\right)^{x+1} + 1$



Match the following functions with their correct graphs.

Place the appropriate letter in this column.

Solution

$$f(x) = \frac{1}{x-2}$$

C

$$g(x) = \frac{-x}{x-2}$$

D

$$h(x) = \frac{2x}{x-1}$$

A

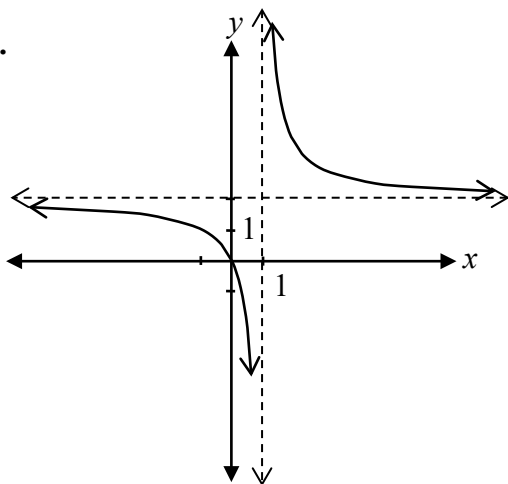
$$k(x) = \frac{2x}{x+1}$$

B

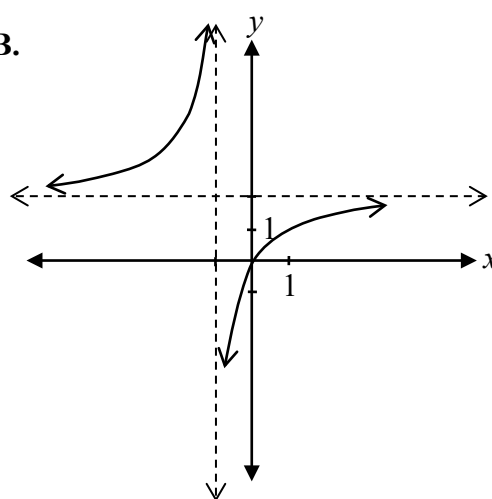
½ mark for each answer

2 marks

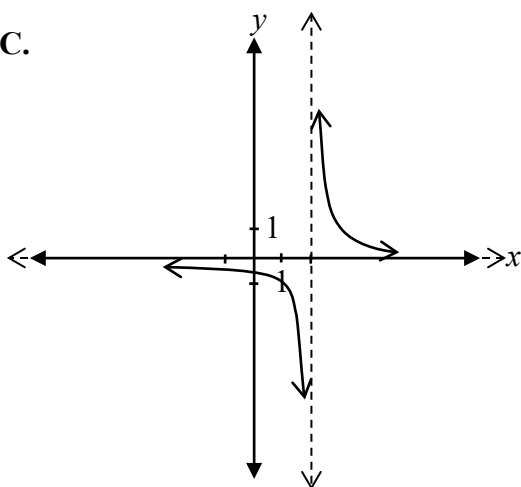
A.



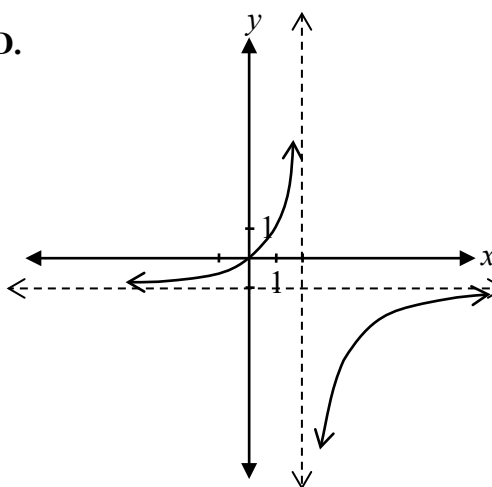
B.



C.



D.



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Evaluate.

$$\sin^2\left(\frac{11\pi}{3}\right)\cot\left(\frac{\pi}{4}\right) + \csc\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{-\sqrt{3}}{2}\right)^2 (1) + (-2)$$

1 mark for $\sin\left(\frac{11\pi}{3}\right)$ (½ mark for quadrant; ½ mark for value)

$$\left(\frac{3}{4}\right)(1) - 2$$

1 mark for $\cot\left(\frac{\pi}{4}\right)$ (½ mark for quadrant; ½ mark for value)

$$\frac{3}{4} - 2$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

$$-\frac{5}{4}$$

3 marks

or

$$-1\frac{1}{4}$$

Exemplar 1

$$\begin{aligned} &= \sin^2\left(\frac{5\pi}{3} + 2\pi\right) \cdot \frac{\cos \frac{\pi}{4}}{\sin \frac{\pi}{4}} \cdot \frac{1}{\sin \frac{11\pi}{6}} \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 \cdot \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \cdot \frac{1}{\frac{1}{2}} \\ &= \frac{3}{4} \cdot 1 \cdot 2 = \frac{3}{2} \end{aligned}$$

2 out of 3

+ ½ mark for value of $\sin\left(\frac{11\pi}{6}\right)$

+ 1 mark for $\cot\left(\frac{\pi}{4}\right)$

+ ½ mark for value of $\csc\left(\frac{11\pi}{6}\right)$

E7 (transcription error in line 1)

Exemplar 2

$$\begin{aligned} &\left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right) + \left(-\frac{1}{2}\right) \\ &\left(-\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right) \\ &\frac{-2\sqrt{3}}{2} \end{aligned}$$

2 out of 3

+ 1 mark for $\sin\left(\frac{11\pi}{6}\right)$

+ 1 mark for $\cot\left(\frac{\pi}{4}\right)$

+ ½ mark for quadrant of $\csc\left(\frac{11\pi}{6}\right)$

– ½ mark for arithmetic error in line 3

E7 (transcription error in line 1)

Exemplar 3

$$\sin^2 \left(2\pi + \frac{5\pi}{3} \right) \cot \left(\frac{\pi}{4} \right) + \csc \left(\pi + \frac{5\pi}{6} \right)$$
$$\left(\frac{\sqrt{3}}{2} \right)^2 \quad \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} + \frac{1}{\frac{1}{2}}$$

$$\left(\frac{3}{4} \right) \quad 1 + 2$$

$$\frac{3}{4} + \frac{2}{1(4)}$$

$$\frac{3}{4} + \frac{8}{4}$$

$$\boxed{\frac{11}{4}}$$

2½ out of 3

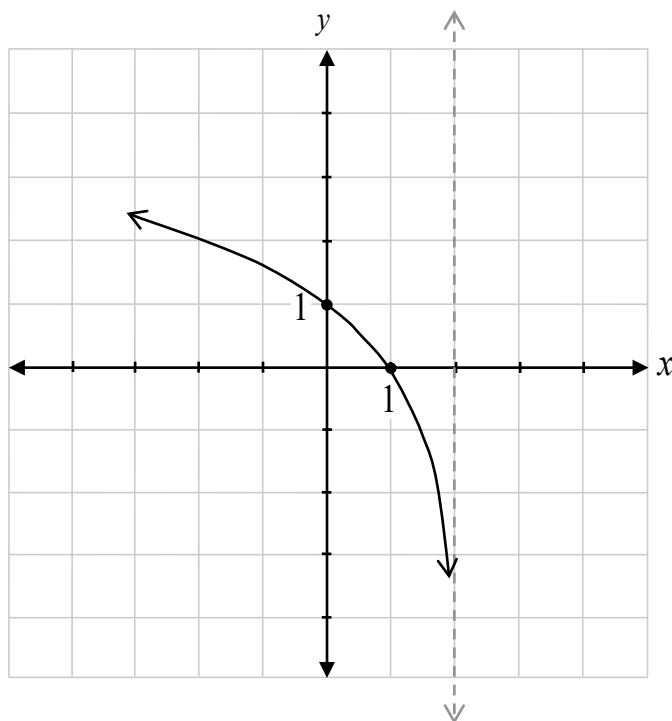
+ 1 mark for value of $\sin \left(\frac{11\pi}{3} \right)$

+ 1 mark for $\cot \left(\frac{\pi}{4} \right)$

+ ½ mark for value of $\csc \left(\frac{11\pi}{6} \right)$

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Sketch the graph of $y = \log_2[-(x-2)]$.

Solution

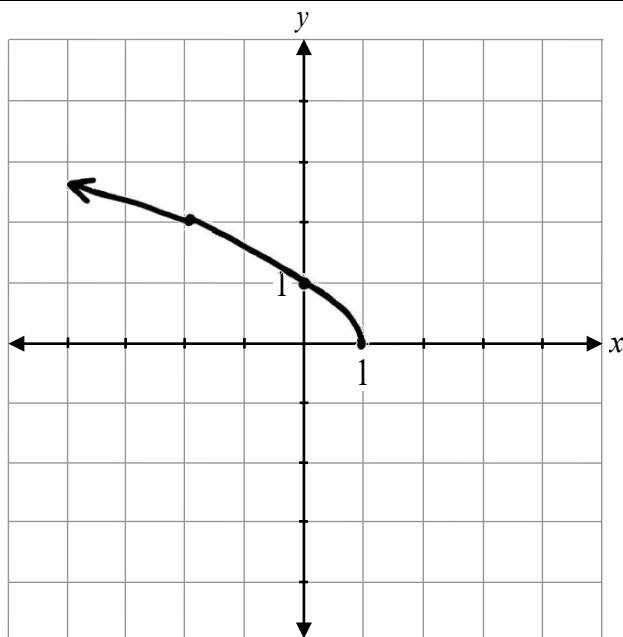
1 mark for shape of a logarithmic function

1 mark for horizontal reflection

1 mark for horizontal translation

3 marks

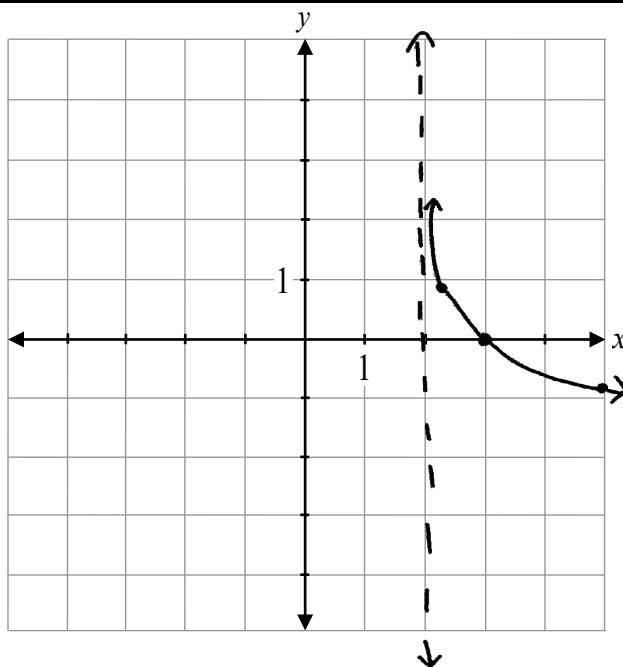
Exemplar 1



2 out of 3

- + 1 mark for horizontal reflection
- + 1 mark for horizontal translation

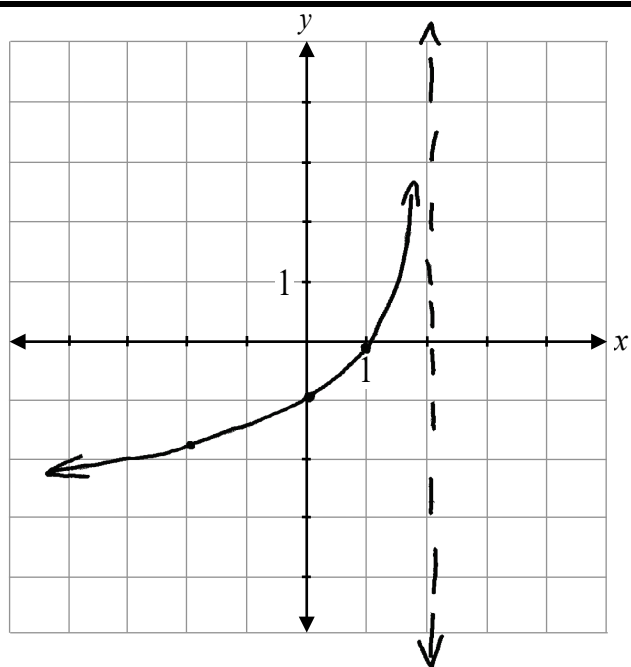
Exemplar 2



1½ out of 3

- + 1 mark for shape of a logarithmic function
- + 1 mark for horizontal translation
- ½ mark for procedural error (incorrect second point)

Exemplar 3



2 out of 3

award full marks

– 1 mark for concept error (added a vertical reflection)

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Justify that ${}_{12}C_7 = {}_{12}C_5$.

Solution

$$\begin{aligned} {}_{12}C_7 &= \frac{12!}{(12-7)!7!} & {}_{12}C_5 &= \frac{12!}{(12-5)!5!} \\ &= \frac{12!}{5!7!} & &= \frac{12!}{7!5!} \end{aligned}$$

$$\frac{12!}{5!7!} = \frac{12!}{7!5!}$$

1 mark

Exemplar 1

If 2 combinations have the same n value and the sum of their r values is equal to the n value, they are equal.

1 out of 1

Exemplar 2

this is because when you add the two numbers, 7 and 5, it equals to 12.

1/2 out of 1

award full marks

– 1/2 mark for lack of clarity in justification

Solve, algebraically.

$$\left(\frac{1}{4}\right)^{3x} = 8^{x-3}$$

Solution

$$\left(2^{-2}\right)^{3x} = \left(2^3\right)^{x-3}$$

$$2^{-6x} = 2^{3x-9}$$

$$-6x = 3x - 9$$

$$-9x = -9$$

$$x = 1$$

1 mark for changing to a common base (½ mark for each side)

½ mark for exponent law

½ mark for equating exponents

2 marks

Exemplar 1

$$3x \log \frac{1}{4} = (x-3) \log 8$$

$$3x \log \frac{1}{4} = x \log 8 - 3 \log 8$$

$$3x \log \frac{1}{4} - x \log 8 = -3 \log 8$$

$$x (3 \log \frac{1}{4} - \log 8) = -3 \log 8$$

$$x = \frac{-3 \log 8}{(3 \log \frac{1}{4} - \log 8)}$$

2 out of 2

award full marks

E1 (final answer not stated)

Exemplar 2

$$\frac{1}{4}^{3x} = \left(\frac{1}{4}\right)^{-2(x-3)}$$

$$3x = -2x + 6$$

$$5x = 6$$

$$x = \frac{6}{5}$$

½ out of 2

+ ½ mark for exponent law

+ ½ mark for equating exponents

– ½ mark for arithmetic error in line 4

E4 (missing brackets but still implied in line 1)

State the domain and range of the function $y = -\sqrt{x-5} + 4$.

Solution

Domain: $\{x|x \geq 5, x \in \mathbb{R}\}$ 1 mark for domain

Range: $\{y|y \leq 4, y \in \mathbb{R}\}$ 1 mark for range

or

Domain: $[5, \infty)$ 1 mark for domain

Range: $(-\infty, 4]$ 1 mark for range

2 marks

Exemplar 1

Domain: $x \geq 5$

Range: $y \geq 4$

1 out of 2

+ 1 mark for domain

Exemplar 2

Domain: $(5, -\infty)$

Range: $(4, -\infty)$

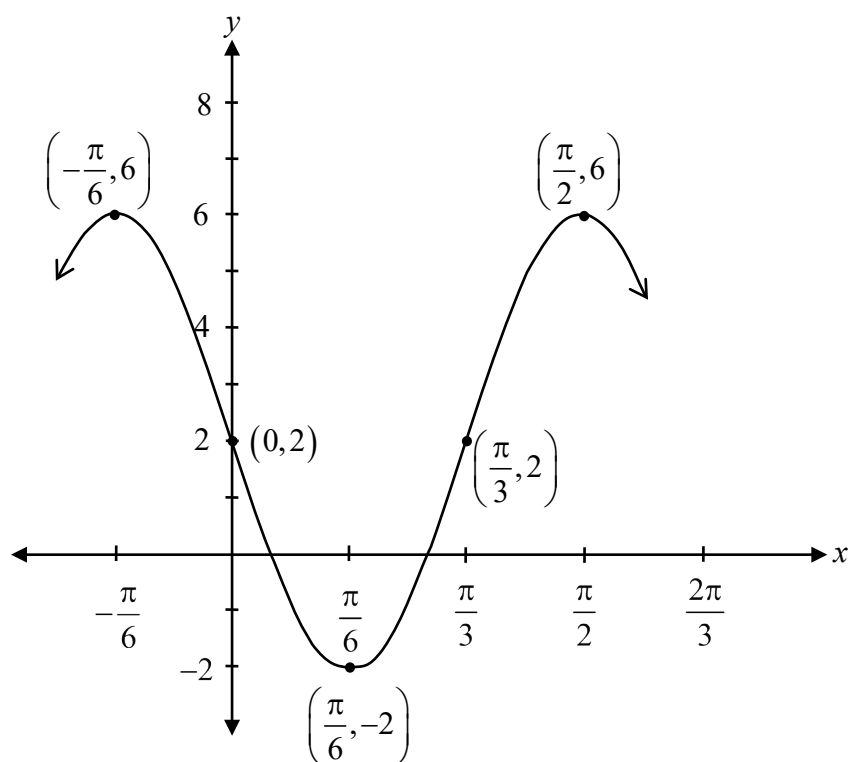
1 out of 2

+ 1 mark for range

E8 (bracket error made when stating range)

E8 (range written in incorrect order)

Sketch at least one period of the graph of $y = 4 \cos \left[3 \left(x + \frac{\pi}{6} \right) \right] + 2$.

Solution

1 mark for shape of a sinusoidal function with correct amplitude

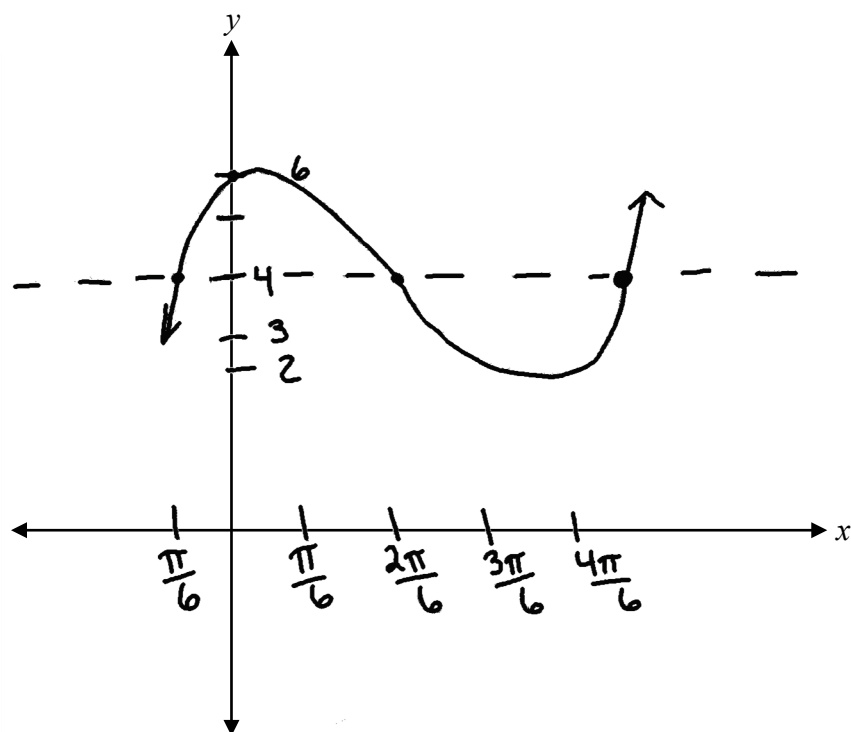
1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Exemplar 1

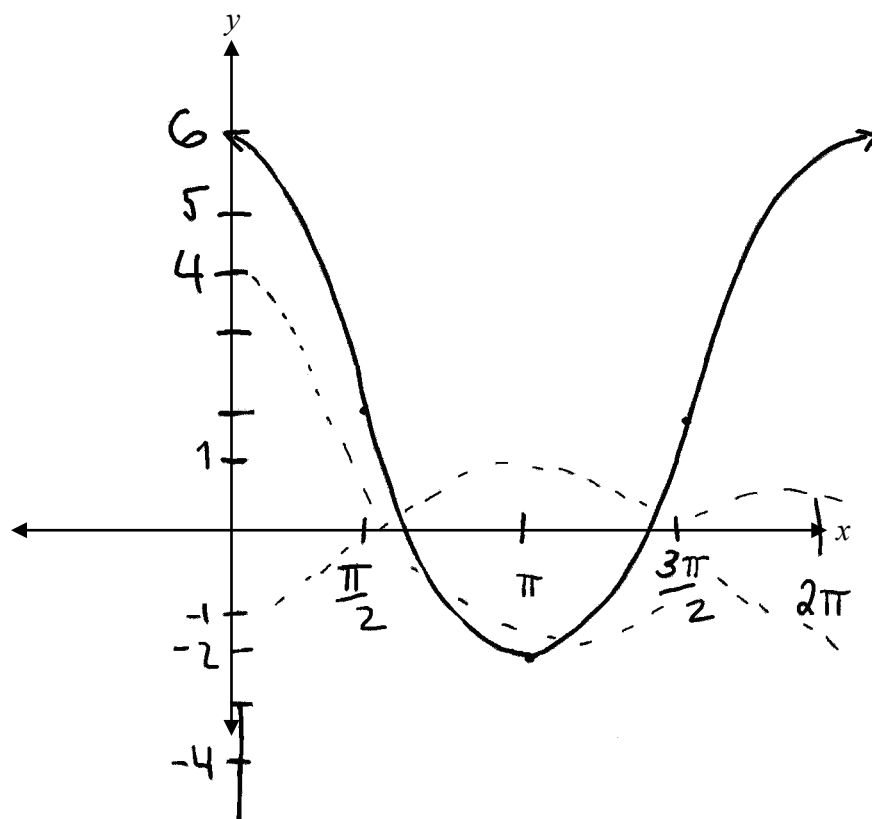


0 out of 4

+ 1 mark for horizontal translation

– 1 mark for concept error (graph of $y = \sin x$)

Exemplar 2



2 out of 4

+ 1 mark for shape of a sinusoidal function with correct amplitude

+ 1 mark for vertical translation

E9 (scale values on y-axis inconsistently spaced)

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Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has zeros at -5 and 1
- all zeros of $p(x)$ have a multiplicity of 2
- the graph of $p(x)$ has a y -intercept at 75

Solution

$$p(x) = a(x+5)^2(x-1)^2$$

1 mark for multiplicity of 2 at $x = -5$ and $x = 1$

$$75 = a(5)^2(-1)^2$$

$$75 = 25a$$

$$a = 3$$

1 mark for consistent value of a

$$p(x) = \underline{3(x+5)^2(x-1)^2}$$

2 marks

Exemplar 1

$$(x+5)^2(x-1)^2$$

$$\begin{array}{r} 25 \quad 1 \\ 25 \\ +50 \\ \hline 75 \end{array}$$

$$p(x) = \underline{(x+5)^2(x-1)^2 + 50}$$

1 out of 2

+ 1 mark for multiplicity of 2 at $x = -5$ and $x = 1$

Exemplar 2

$$p(x) = \underline{(x+5)^2(x-1)^2 + 75}$$

1 out of 2

+ 1 mark for multiplicity of 2 at $x = -5$ and $x = 1$

Exemplar 3

$$p(x) = \underline{3(x-5)^2(x+1)^2}$$

1½ out of 2

award full marks

– ½ mark procedural error (incorrect zeros)

Vitaly was asked to solve the exponential equation $5^{2x+7} = 3 \cdot 7^x$.

Vitaly's work:

$$\begin{aligned}\log(5^{2x+7}) &= \log(3 \cdot 7^x) \\ (2x+7)\log 5 &= \log(21^x) \\ 2x\log 5 + 7\log 5 &= x\log 21 \\ 7\log 5 &= x\log 21 - 2x\log 5 \\ 7\log 5 &= x(\log 21 - 2\log 5) \\ x &= \frac{7\log 5}{\log 21 - 2\log 5}\end{aligned}$$

Describe his error.

Solution

In step 2, Vitaly incorrectly simplified $3 \cdot 7^x$ to 21^x .
He should have applied the product law of logarithms.

1 mark

Exemplar 1

You cannot multiply an integer when it
is to any power.

½ out of 1

award full marks

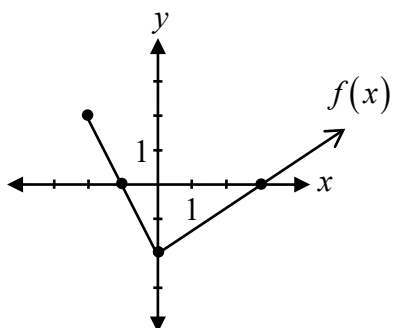
– ½ mark for lack of clarity in description

Exemplar 2

On the second line, he multiplied $3 \cdot 7^x$. She cannot do that
since BEMAS rules states exponents before multiplying.

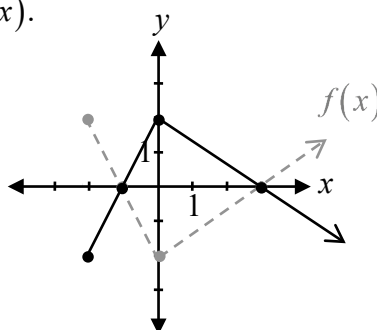
1 out of 1

Given the graph of $f(x)$,



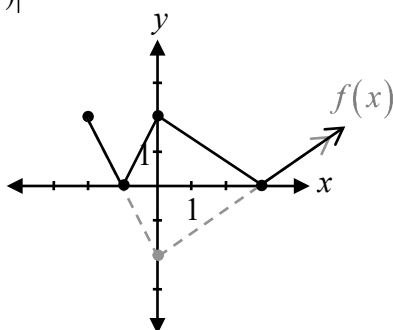
Solution

a) sketch the graph of $g(x) = -f(x)$.



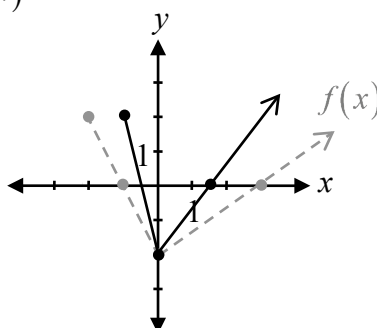
1 mark

b) sketch the graph of $k(x) = |f(x)|$.



1 mark

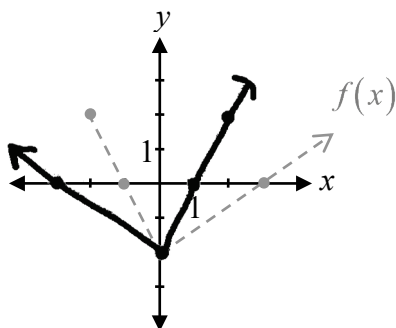
c) sketch the graph of $r(x) = f(2x)$.



1 mark

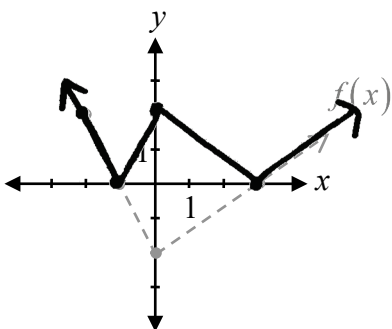
Exemplar 1

a)



0 out of 1

b)

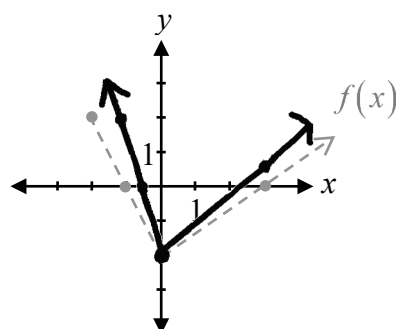


1 out of 1

award full marks

E9 (endpoint incorrect)

c)



½ out of 1

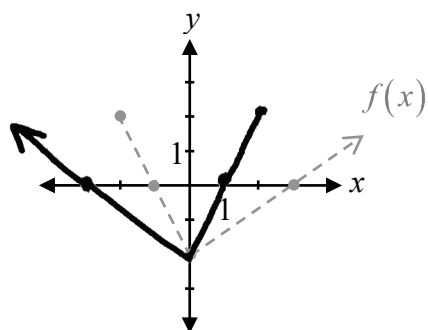
award full marks

– ½ mark for procedural error (one incorrect point)

E9 (endpoint incorrect)

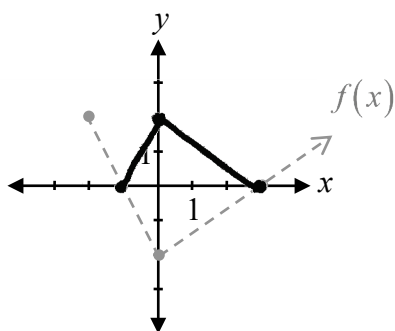
Exemplar 2

a)



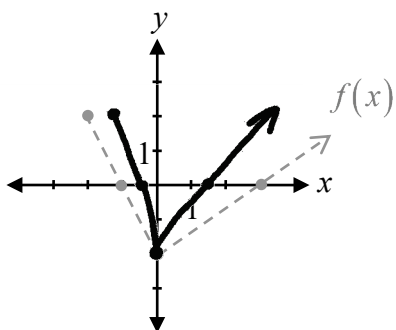
0 out of 1

b)



0 out of 1

c)



1 out of 1

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Determine the exact value of $\sin\left(\frac{17\pi}{12}\right)$.

Solution

$$\sin\left(\frac{17\pi}{12}\right) = \sin\left(\frac{\pi}{6} + \frac{5\pi}{4}\right)$$

$$= \sin\left(\frac{\pi}{6}\right)\cos\left(\frac{5\pi}{4}\right) + \cos\left(\frac{\pi}{6}\right)\sin\left(\frac{5\pi}{4}\right)$$

1 mark for substitution into correct identity

$$= \left(\frac{1}{2}\right)\left(\frac{-\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{-\sqrt{2}}{2}\right)$$

2 marks for exact values (½ mark for each)

$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

3 marks

Note:

- Other combinations are possible.

Exemplar 1

$$\sin\left(\frac{9\pi}{12}\right) + \sin\left[\frac{8\pi}{12}\right]$$

$$\sin\left[\frac{3\pi}{4}\right] + \sin\left[\frac{2\pi}{3}\right]$$

$$\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}$$

$$\boxed{\frac{\sqrt{2} + \sqrt{3}}{2}}$$

1 out of 3

+ 1 mark for two exact values

Exemplar 2

$$\sin\left(\frac{14\pi}{12} + \frac{3\pi}{12}\right)$$

$$\sin\left(\frac{7\pi}{6} + \frac{\pi}{4}\right) = \sin\frac{7\pi}{6} \cos\frac{\pi}{4} + \cos\frac{7\pi}{6} \sin\frac{\pi}{4}$$

$$= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} + -\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{2}}{4} + -\frac{\sqrt{3} \cdot \sqrt{2}}{4}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

2½ out of 3

+ 1 mark for substitution into correct identity

+ 1½ marks for three exact values

Given $f(x) = \sqrt{x+3}$ and $g(x) = x-8$,

a) determine the value of $g(f(1))$.

b) justify that the value of $(f-g)(-5)$ is undefined.

Solution

a) $f(1) = 2$ $\frac{1}{2}$ mark for the value of $f(1)$

$$g(2) = -6$$

$g(f(1)) = -6$ $\frac{1}{2}$ mark for consistent value of $g(f(1))$

1 mark

b) The domain of $f(x)$ is $x \geq -3$; $x = -5$ is outside the domain.

or

$$\begin{aligned} f(-5) &= \sqrt{-5+3} \\ &= \sqrt{-2} \end{aligned}$$

It is not possible to evaluate the square root of a negative radicand, therefore $f(-5)$ is undefined.

1 mark

Exemplar 1

a)

$$\begin{aligned} f(1) &= \sqrt{1+3} \\ &= \sqrt{4} \\ &= 2 \end{aligned}$$

$$\begin{aligned} g(2) &= 2-8 \\ &= -6 \end{aligned}$$

$$\boxed{g(f(1)) = -6}$$

1 out of 1

b)

$$\begin{aligned} (f-g)(x) &= \sqrt{(x-8)+3} \\ &= \sqrt{x-8+3} \\ &= \sqrt{x-5} \end{aligned}$$

$$(f-g)(-5) = \sqrt{-5-5}$$

$$= \sqrt{-10}$$

undefined

Can't square

root a negative

value.

1/2 out of 1

award full marks

– 1/2 mark for procedural error (composition of function)

Exemplar 2

a)

$$\begin{array}{c} g(2) \\ -6 \end{array}$$

1 out of 1

b)

we can't $\sqrt{\quad}$ root a negative number.

½ out of 1

award full marks

– ½ mark for lack of clarity in justification

Exemplar 3

a)

$$f(1) = \sqrt{1+3} = \sqrt{4} = 2$$

$$g(f(1)) = 2 - 8$$

$$= \boxed{-6}$$

1 out of 1

b)

$$\sqrt{-5+3} - (5-8)$$

$$= \sqrt{-2} + 3$$

undefined because of the negative
in the square root.

1 out of 1

award full marks

E7 (transcription error)

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Solve, algebraically.

$$48 \binom{n}{4} = {}_{(n+1)}P_4$$

Solution

$$48 \left(\frac{n!}{(n-4)!4!} \right) = \frac{(n+1)!}{((n+1)-4)!}$$

1 mark for substitution into equation (½ mark for each side)

$$48 \left(\frac{n!}{(n-4)!4!} \right) = \frac{(n+1)!}{(n-3)!}$$

$$\frac{48}{4!} = \frac{(n-4)!(n+1)!}{(n-3)!n!}$$

$$\frac{48}{24} = \frac{\cancel{(n-4)!} (n+1) \cancel{(n)!}}{(n-3) \cancel{(n-4)!} \cancel{(n)!}}$$

1 mark for factorial expansion
½ mark for simplification of factorials

$$2 = \frac{n+1}{n-3}$$

$$2n - 6 = n + 1$$

$$n = 7$$

½ mark for solving for n

3 marks

Exemplar 1

$$48 \left(\frac{n!}{4!(n-4)!} \right) = \frac{(n+1)!}{4!(n-3)!}$$

$$48 \frac{\cancel{n}(\cancel{n-1})(\cancel{n-2})(\cancel{n-3})(n-4)!}{4!(\cancel{n-4})!} = \frac{(n+1)(\cancel{n})(\cancel{n-1})(\cancel{n-2})(\cancel{n-3})!}{4!(\cancel{n-3})!}$$

$$48(n-3) = n+1$$

$$48n - 144 = n+1$$

$$47n = 145$$

$$\frac{47}{145} = n$$

2 out of 3

- + ½ mark for substitution into equation
- + 1 mark for factorial expansion
- + ½ mark for simplification of factorials

Exemplar 2

$$48 \left(\frac{n!}{4!(n-4)!} \right) = \frac{n+1!}{((n+1)-4)!}$$

1 out of 3

- + 1 mark for substitution into equation
- E7 (notation errors)

Exemplar 3

$$48 \left(\frac{n!}{4!(n-4)!} \right) = \frac{n+1!}{(n+1-4)!}$$

$$\frac{48n!}{24(n-4)!} = \frac{n+1!}{(n-3)!}$$

$$\frac{48\cancel{n!}}{24(n-4)\cancel{!}} = \frac{(n+1)(\cancel{n!})}{(n-3)(n-4)\cancel{!}}$$

$$\frac{48}{24} = n+1$$

$$2 = n+1$$

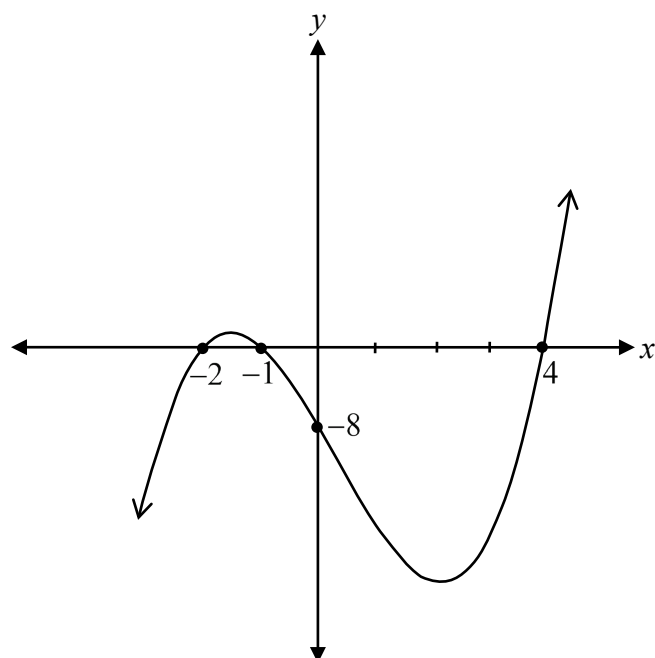
$$\boxed{n=1}$$

2½ out of 3

- + 1 mark for substitution into equation
 - + 1 mark for factorial expansion
 - + ½ mark for simplification of factorials
- E4 (missing brackets but still implied in lines 1 and 2)

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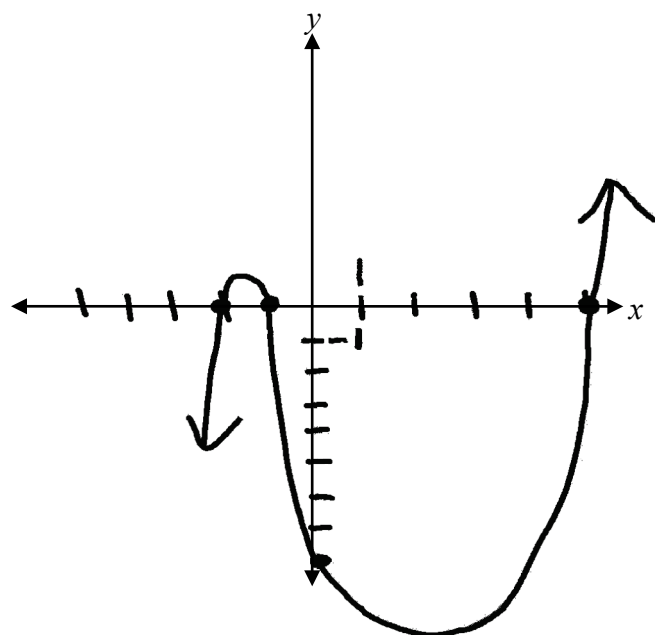
Sketch the graph of $p(x) = (x+1)(x+2)(x-4)$.

Solution

1 mark for x -intercepts
 $\frac{1}{2}$ mark for y -intercept
 $\frac{1}{2}$ mark for end behaviour

2 marks

Exemplar 1

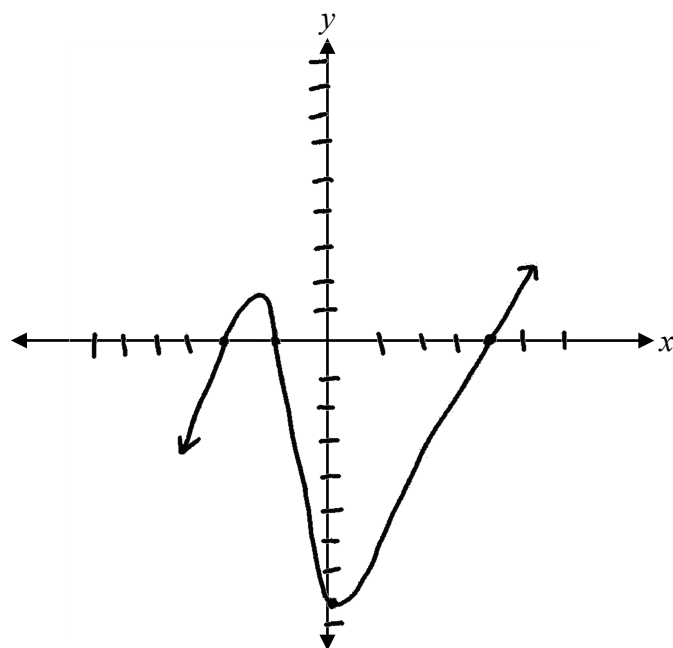


1½ out of 2

award full marks

– ½ mark for procedural error (one incorrect x -intercept)

Exemplar 2

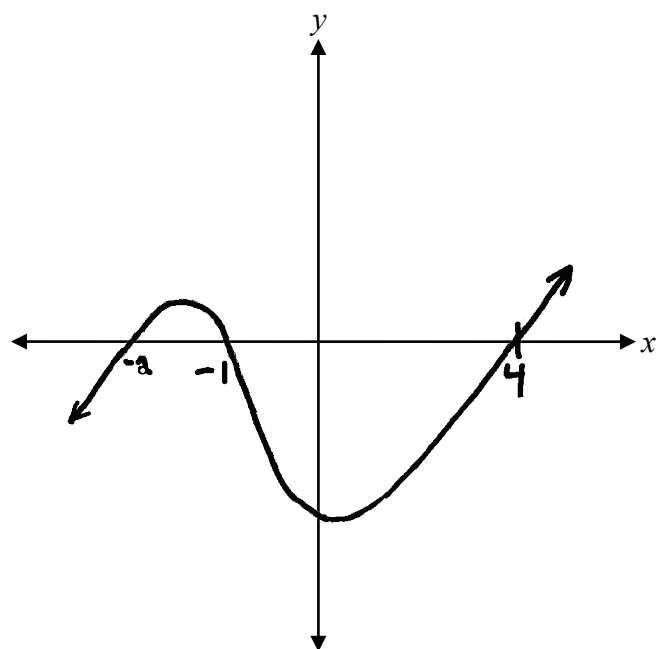


2 out of 2

award full marks

E9 (scale values on axes not indicated)

Exemplar 3



1½ out of 2

+ 1 mark for x -intercepts

+ ½ mark for end behaviour

E9 (scale values inconsistently spaced on the x -axis)

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Evaluate.

$$\log_5(\log_2 32)$$

Solution

$$\log_5 5$$

$\frac{1}{2}$ mark for evaluating $\log_2 32$

$$1$$

$\frac{1}{2}$ mark for consistent value

1 mark

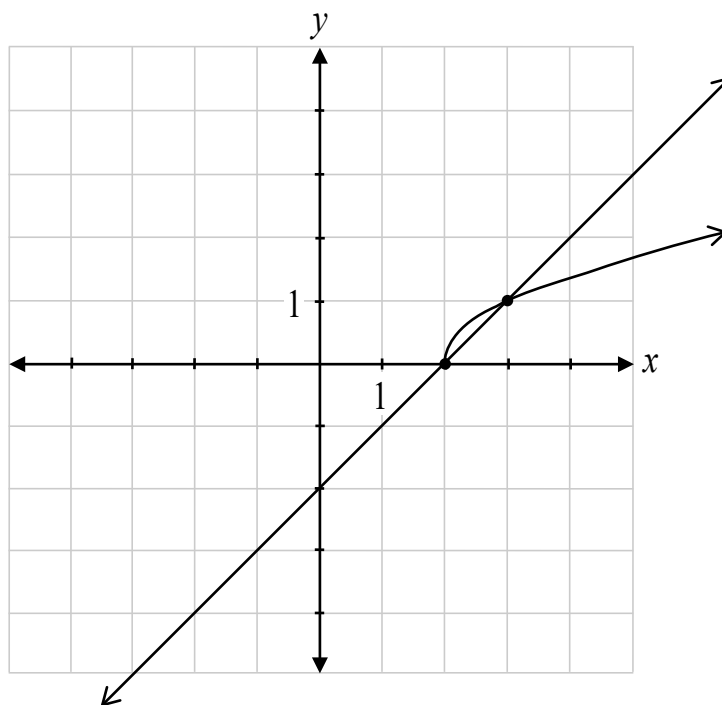
Exemplar 1

$$\begin{aligned} &\log_5(\log_2 2^5) \\ &\log_5(5 \log_2 2^3) \\ &\log_5(5) \end{aligned}$$

½ out of 1

+ ½ mark for evaluating $\log_2 32$

Describe how to use the graph to solve the equation $\sqrt{x-2} = x-2$.

**Solution**

State the x -coordinates of the points of intersection.

1 mark

Exemplar 1

To solve the equation by using the graphs find where the two functions intersect, which is at $(3, 1)$, so

$$x = 3.$$

½ out of 1

award full marks

– ½ mark for lack of clarity in description (only including one solution)

Exemplar 2

where the graphs cross
 $\sqrt{x-2} = x-2$

½ out of 1

award full marks

– ½ mark for lack of clarity in description

Exemplar 3

look @ the 2 values to determine the answer.

0 out of 1

Verify, by substitution, that the equation $\frac{\sec \theta}{\tan \theta + \cot \theta} = \sin 2\theta$ is true for $\theta = \frac{\pi}{3}$.

Solution

Left-Hand Side	Right-Hand Side
$\frac{\sec \frac{\pi}{3}}{\tan \frac{\pi}{3} + \cot \frac{\pi}{3}}$ $\frac{2}{\sqrt{3} + \frac{1}{\sqrt{3}}}$ $\frac{2}{\frac{3+1}{\sqrt{3}}}$ $\frac{2\sqrt{3}}{4}$ $\frac{\sqrt{3}}{2}$	$\sin \frac{2\pi}{3}$ $\frac{\sqrt{3}}{2}$ ½ mark for substitution 2 marks for exact values (½ mark for each) ½ mark for simplification
LHS = RHS	

3 marks

Exemplar 1

Left-Hand Side	Right-Hand Side
$\frac{\sec\left(\frac{\pi}{3}\right)}{\tan\left(\frac{\pi}{3}\right) + \cot\left(\frac{\pi}{3}\right)}$ $\frac{2}{\frac{\sqrt{3}}{1} + \frac{1}{\sqrt{3}}}$ $\sqrt{3}$	$\sin 2\left(\frac{\pi}{3}\right)$ $\cancel{2}\left(\frac{\sqrt{3}}{\cancel{2}}\right)$ $\sqrt{3}$
$L.S. \neq R.S.$	

2 out of 3

- + ½ mark for substitution
- + 2 marks for exact values
- ½ mark for procedural error in line 2 on RHS

Exemplar 2

Left-Hand Side	Right-Hand Side
$\frac{\sec \frac{\pi}{3}}{\tan \frac{\pi}{3} \cot \frac{\pi}{3}}$ $\frac{2}{\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right)\left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)}$ $\frac{2}{\left(\frac{\sqrt{3} \cdot 2}{2 \cdot 1}\right)\left(\frac{1 \cdot 2}{2 \cdot \sqrt{3}}\right)}$ $\frac{2}{\left(\frac{2\sqrt{3}}{2}\right)\left(\frac{2}{2\sqrt{3}}\right)}$ 2	$\sin 2\left(\frac{\pi}{3}\right)$ $2 \sin \frac{\pi}{3} \cos \frac{\pi}{3}$ $2 \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)$ $\frac{2}{1} \left(\frac{\sqrt{3}}{4}\right)$ $\frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$

3 out of 3

award full marks

E7 (transcription error in line 1)

Exemplar 3

Left-Hand Side	Right-Hand Side
$\frac{\sec \theta}{\tan \theta + \cot \theta}$ $\frac{1}{\cos \frac{\pi}{3}}$ $\frac{\frac{\sin \frac{\pi}{3}}{\cos \frac{\pi}{3}} + \frac{\cos \frac{\pi}{3}}{\sin \frac{\pi}{3}}}{1}$ $= \frac{1}{\cos \frac{\pi}{3}}$ $\frac{\frac{1}{\frac{1}{2}} + \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}}{1}$ $= \frac{1}{\frac{1}{2} + \frac{\sqrt{3}}{2}}$ $= \frac{\sqrt{3}}{2} \left(\frac{4}{\sqrt{3}} \right)$	$\sin 2\theta$ $2 \sin \left(\frac{\pi}{3} \right) \cos \left(\frac{\pi}{3} \right)$ $= 2 \left(\frac{1}{2} \right) \left(\frac{\sqrt{3}}{2} \right)$ $= 1 \left(\frac{\sqrt{3}}{2} \right)$

1½ out of 3

+ ½ mark for substitution

+ 2 marks for exact values

– 1 mark for concept error (switching $\cos \frac{\pi}{3}$ and $\sin \frac{\pi}{3}$)

Describe the transformation(s) to the graph of $y = 3^x$ which results in a graph with a y -intercept of -2 .

Solution

A vertical translation down 3 units.

1 mark

Note:

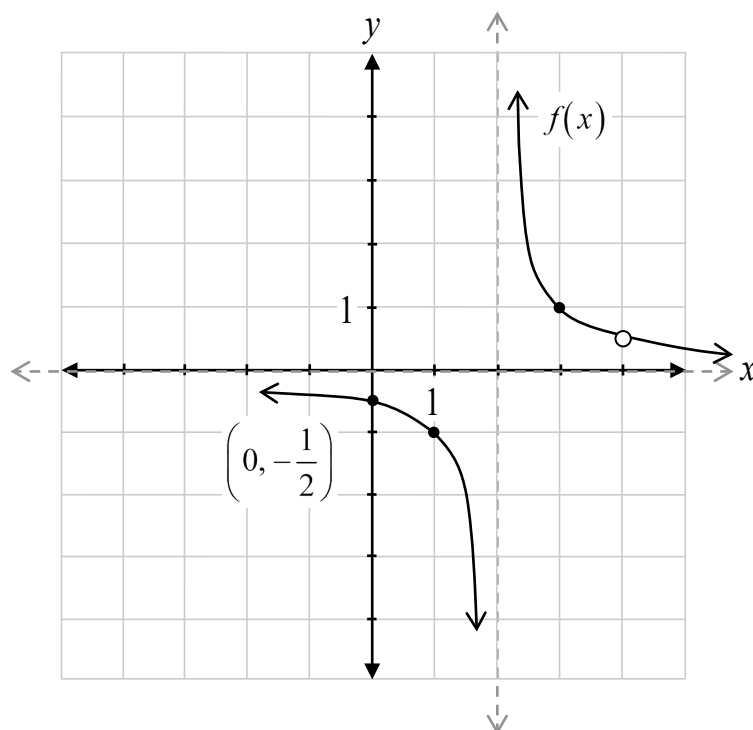
- Other answers are possible.

Exemplar 1

The transformation is a reflection about the x -axis and a vertical stretch of 2.

1 out of 1

Determine the equation of the rational function represented by the graph.

**Solution**

$$f(x) = \frac{x-4}{(x-2)(x-4)}$$

1 mark for factor of $(x-2)$ in denominator

1 mark for factor of $(x-4)$ in both numerator and denominator

2 marks

Exemplar 1

$$f(x) = \frac{x}{x-2}$$

1 out of 2

+ 1 mark for factor of $(x-2)$ in denominator

Exemplar 2

$$f(x) = \frac{x-2}{x-4}$$

0 out of 2

Exemplar 3

$$f(x) = \frac{1}{(x-2)}, x \neq 4$$

2 out of 2

Exemplar 4

$$f(x) = \frac{(x-4)(x)}{(x-2)(x-4)}$$

1½ out of 2

award full marks

– ½ mark for procedural error (incorrect horizontal asymptote)

Justify that $f(x) = 3x + 2$ and $g(x) = \frac{x-3}{2}$ are not inverses of each other.

Solution**Method 1**

$$\text{Let } y = f(x)$$

$$y = 3x + 2$$

$$\text{Let } y = f^{-1}(x)$$

$$x = 3y + 2$$

1 mark for switching x and y

$$\frac{x-2}{3} = y$$

$$\frac{x-2}{3} = f^{-1}(x)$$

$$g(x) \neq f^{-1}(x)$$

1 mark for justification

2 marks**Method 2**

If they are inverses, then $g(f(x)) = x$, and $f(g(x)) = x$.

$$\begin{aligned} g(3x+2) &= \frac{3x+2-3}{2} \\ &= \frac{3x-1}{2} \end{aligned}$$

1 mark for $g(f(x))$ or $f(g(x))$

Since $g(f(x)) \neq x$, they are not inverses of each other.

1 mark for justification

2 marks

Exemplar 1

$$x = 3y + 2$$

$$x - 2 = 3y$$

$$\frac{x-2}{3} = y$$

$$f^{-1}(x) = \frac{x-2}{3}$$

$$x = \frac{y-3}{2}$$

$$2x = y - 3$$

$$2x + 3 = y$$

$$g^{-1}(x) = 2x + 3$$

not inverse

1 out of 2

+ 1 mark for switching x and y

Exemplar 2

$$\text{let } y = f(x)$$

$$y = 3x + 2$$

$$\begin{array}{r} x = 3y + 2 \\ -2 \quad -2 \end{array}$$

$$\frac{x-2}{3} = \frac{3y}{3}$$

$$\frac{x-2}{3} = y$$

1 out of 2

+ 1 mark for switching x and y

The terminal arm of an angle, θ , in standard position passes through the point $(-3, 6)$ and intersects the unit circle at the point $P(\theta)$. Determine the coordinates of the point $P(\theta)$.

Solution

$$x^2 + y^2 = r^2$$

$$(-3)^2 + (6)^2 = r^2 \quad \frac{1}{2} \text{ mark for substitution}$$

$$9 + 36 = r^2$$

$$45 = r^2$$

$$\sqrt{45} = r \quad \frac{1}{2} \text{ mark for solving for } r$$

$$P(\theta) = \left(\frac{-3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right) \quad \begin{array}{l} 1 \text{ mark for the coordinates of } P(\theta) \\ (\frac{1}{2} \text{ mark for } x\text{-coordinate; } \frac{1}{2} \text{ mark for } y\text{-coordinate}) \end{array}$$

or

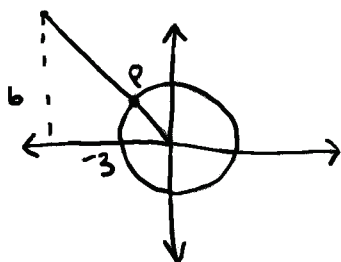
$$P(\theta) = \left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

or

$$P(\theta) = \left(\frac{-\sqrt{5}}{5}, \frac{2\sqrt{5}}{5} \right)$$

2 marks

Exemplar 1



$$\sqrt{9 + 36} = c$$

$$\sqrt{45} = c$$

$$P = \left(-\frac{3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right)$$

2 out of 2

award full marks

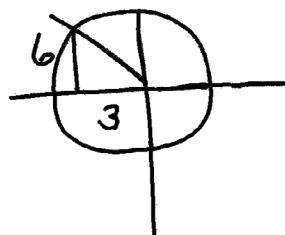
E7 (transcription error)

Exemplar 2

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 9 + 36 &= 45 \end{aligned}$$

$$\cos \theta = \frac{3}{\sqrt{45}}$$

$$\sin \theta = \frac{6}{\sqrt{45}}$$



$$P = \left(\frac{3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right)$$

1½ out of 2

+ ½ mark for substitution

+ ½ mark for solving for r

+ ½ mark for y -coordinate

Determine the exact value of $2\sin\theta\cos\theta$ if $\theta = \frac{\pi}{12}$.

Solution

$2\sin\theta\cos\theta = \sin 2\theta$ 1 mark for substitution into correct identity

$$= \sin\left(2\left(\frac{\pi}{12}\right)\right)$$

$$= \sin\left(\frac{\pi}{6}\right)$$

$$= \frac{1}{2}$$

1 mark for consistent value

2 marks

Exemplar 1

$$2 \sin \frac{\pi}{12} \cdot \cos \frac{\pi}{12}$$

0 out of 2

Exemplar 2

$$= 2 \sin \frac{\pi}{12} \cos \frac{\pi}{12}$$

$$= \sin 2 \frac{\pi}{12}$$

$$= \sin \frac{2\pi}{12}$$

$$= \sin \frac{\pi}{6}$$

$$= \frac{\sqrt{3}}{2}$$

1 out of 2

+ 1 mark for substitution into correct identity

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Appendices

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Appendix A

MARKING GUIDELINES

Errors that are conceptually related to the learning outcomes associated with the question will result in a 1 mark deduction.

Each time a student makes one of the following errors, a $\frac{1}{2}$ mark deduction will apply:

- arithmetic error
- procedural error
- terminology error in explanation
- lack of clarity in explanation, description, or justification
- incorrect shape of graph (only when marks are not allocated for shape)

Communication Errors

The following errors, which are not conceptually related to the learning outcomes associated with the question, may result in a $\frac{1}{2}$ mark deduction and will be tracked on the *Answer/Scoring Sheet*.

E1 final answer	▪ answer given as a complex fraction ▪ final answer not stated ▪ impossible solution(s) not rejected in final answer and/or in step leading to final answer
E2 equation/expression	▪ changing an equation to an expression or vice versa ▪ equating the two sides when proving an identity
E3 variables	▪ variable omitted in an equation or identity ▪ variables introduced without being defined
E4 brackets	▪ "$\sin x^2$" written instead of "$\sin^2 x$" ▪ missing brackets but still implied
E5 units	▪ units of measure omitted in final answer ▪ incorrect units of measure ▪ answer stated in degrees instead of radians or vice versa
E6 rounding	▪ rounding error ▪ rounding too early
E7 notation/transcription	▪ notation error ▪ transcription error
E8 domain/range	▪ answer outside the given domain ▪ bracket error made when stating domain or range ▪ domain or range written in incorrect order
E9 graphing	▪ endpoints or arrowheads omitted or incorrect ▪ scale values on axes not indicated or inconsistently spaced ▪ coordinate points labelled incorrectly
E10 asymptotes	▪ asymptotes drawn as solid lines ▪ asymptotes omitted but still implied ▪ graph crosses or curls away from asymptotes

Appendix B

IRREGULARITIES IN PROVINCIAL TESTS

A GUIDE FOR LOCAL MARKING

During the marking of provincial tests, irregularities are occasionally encountered in test booklets. The following list provides examples of irregularities for which an *Irregular Test Booklet Report* should be completed and sent to the department:

- completely different penmanship in the same test booklet
- incoherent work with correct answers
- notes from a teacher indicating how he or she has assisted a student during test administration
- student offering that he or she received assistance on a question from a teacher
- student submitting work on unauthorized paper
- evidence of cheating or plagiarism
- disturbing or offensive content
- no responses provided by the student or only incorrect responses ("0")

Student comments or responses indicating that the student may be at personal risk of being harmed or of harming others are personal safety issues. This type of student response requires an immediate and appropriate follow-up at the school level. In this case, please ensure the department is made aware that follow-up has taken place by completing an *Irregular Test Booklet Report*.

Except in the case of cheating or plagiarism where the result is a provincial test mark of 0%, it is the responsibility of the division or the school to determine how they will proceed with irregularities. Once an irregularity has been confirmed, the marker prepares an *Irregular Test Booklet Report* documenting the situation, the people contacted, and the follow-up. The original copy of this report is to be retained by the local jurisdiction and a copy is to be sent to the department along with the test materials.

Irregular Test Booklet Report

Test: _____

Date marked: _____

Booklet No.: _____

Problem(s) noted: _____

Question(s) affected: _____

Action taken or rationale for assigning marks: _____

Follow-up: _____

Decision: _____

Marker's Signature: _____

Principal's Signature: _____

For Department Use Only—After Marking Complete

Consultant: _____

Date: _____

Appendix C

Table of Questions by Unit and Learning Outcome

Unit A: Transformations of Functions		
Question	Learning Outcome	Mark
7	R2, R3, R5	3
9	R4, R5	2
16	R1	1
17	R2, R6	1
35a)	R5	1
35b)	R1	1
35c)	R3	1
37a)	R1	1
37b)	R1	1
45	R6	2
Unit B: Trigonometric Functions		
Question	Learning Outcome	Mark
5	T1	2
14	T1	1
15	T4	1
19	T3	1
27	T3	3
32	T4	4
46	T2	2
47	T3	1
Unit C: Binomial Theorem		
Question	Learning Outcome	Mark
1	P1	1
4	P4	3
22	P3	1
23	P2	1
24	P4	1
29	P3	1
38	P2, P3	3
Unit D: Polynomial Functions		
Question	Learning Outcome	Mark
12	R11	3
13	R12	1
18	R12	1
33	R12	2
39	R12	2

Unit E: Trigonometric Equations and Identities		
Question	Learning Outcome	Mark
3	T5	3
11	T6	3
20	T6	1
36	T6	3
42	T5	3
47	T6	1
Unit F: Exponents and Logarithms		
Question	Learning Outcome	Mark
2	R10	2
10	R8	3
21	R10	1
25	R9	1
28	R9	3
30	R10	2
34	R8	1
40	R7	1
43	R9	1
Unit G: Radicals and Rationals		
Question	Learning Outcome	Mark
6	R13	3
8	R14	1
9	R13	1
26	R14	2
31	R13	2
41	R13	1
44	R14	2